

# Chapter 13. OTP

RP2350 provides 8 kB of one-time programmable storage (OTP), which holds:

- Preprogrammed per-device information, such as unique device identifier and oscillator trim values
- Security configuration such as debug disable and secure boot enable
- Public key fingerprints for secure boot
- Symmetric keys for decryption of flash contents into SRAM
- Configuration for the USB bootloader, such as customising VID/PID and descriptors
- Bootable software images, for low-cost flashless applications or custom bootloaders
- Any other user-defined data, such as per-device personalisation values

For the full listing of predefined OTP contents, see [Section 13.9](#).

OTP is physically an array of 4096 rows of 24 bits each. You can directly access these 24-bit values, but there is also hardware support for storing 16 bits of data in each row, with 6 bits of Hamming ECC protection and 2 bits of bit polarity reversal protection, yielding an ECC data capacity of 8192 bytes.

On a blank device, the OTP contents is all zeroes, except for some basic device information pre-programmed during manufacturing test. Each bit can be *irreversibly* programmed from zero to one. To program the OTP contents:

- directly access the registers using the SBPI bridge
- call the bootrom `otp_access` API ([Section 5.4.8.21](#))
- use the PICOB00T interface of the USB bootloader ([Section 5.6](#))

RP2350 enforces page-based permissions on OTP to partition Secure from Non-secure data and to ensure that contents that should not change *do not* change. The OTP address space is logically partitioned into 64 pages, each 64 rows in size, for a total of 128 bytes of ECC data per page. Pages initially have full read-write permissions, but can be restricted to read-only or inaccessible for each of Secure, Non-secure and bootloader access.

The page permissions themselves are stored in OTP. Locking pages in this way is an irreversible operation, referred to as *hard locking*. The hardware also supports *soft locking*, where a page's permissions are further restricted by writing to the relevant register in `SW_LOCK0` through `SW_LOCK63`; this restriction remains in effect until the next OTP reset. Resetting the OTP block also resets the processors, so soft locking can be used to restrict the availability of sensitive content like decryption keys to early boot stages.

OTP access keys ([Section 13.5.2](#)) provide an additional layer of protection. A fixed challenge is written to a write-only OTP area. Pages registered to that key require the key to be entered to a write-only register in order to open read or write access. This supports configuration data that can be accessed or edited by the board manufacturer, but not by general firmware running on the device.

## 13.1. OTP Address Map

The OTP hardware resides in a 128 kB region starting at `0x40120000` (OTP\_BASE in the SDK). Bit 16 of the address is used to select either the OTP control registers, in the lower 64 kB, or one of the OTP read data aliases, in the upper 64 kB of this space.

The OTP control registers ([Section 13.8](#)) are aliased at 4 kB intervals to implement the usual set, clear, and XOR atomic write aliases described in [Section 2.1.3](#).

The read data region starting at `0x40130000` divides further into four aliases:

- `0x40130000`, `OTP_DATA_BASE`: ECC read alias. A 32-bit read returns the ECC-corrected data for two neighbouring rows, or all-ones on permission failure. Only the first 8 kB is populated.

- `0x40138000`, `OTP_DATA_GUARDED_BASE`: ECC guarded read alias. Successful reads return the same data as `OTP_DATA_BASE`. Only the first 8 kB is populated.
- `0x40134000`, `OTP_DATA_RAW_BASE`: raw read alias. A 32-bit read directly returns the 24-bit contents of a single row, with zeroes in the eight MSBs, or returns all-ones on permission failure.
- `0x4013c000`, `OTP_DATA_RAW_GUARDED_BASE`: raw, guarded read alias. Successful reads return the same data as `OTP_DATA_RAW_BASE`.

Bit 14 of the address selects ECC (0) vs raw (1). Bit 15 of the address selects unguarded (0) vs guarded (1) access. Guarded reads return the same data as unguarded reads, but perform additional hardware consistency checks and return bus faults on permission failure. For more information, see [Section 13.1.1](#).

#### ! IMPORTANT

The read data regions starting at `0x40130000` are accessible only when `USR.DCTRL` is set, otherwise all reads return a bus error response. This bit is clear when the OTP is being programmed via the SBPI bridge.

Writing to the read data aliases is not a valid operation, and will always return a bus fault. The OTP is programmed by the SBPI bridge, which is used internally by the bootrom `otp_access` API, [Section 5.4.8.21](#).

### 13.1.1. Guarded Reads

Reads through the guarded aliases differ from unguarded reads in the following ways:

- Permission failures return bus faults rather than a bit pattern of all-ones.
- Uncorrectable ECC errors return a bus fault if detected.
- Guarded reads perform an additional hardware consistency check to detect power transients. If this check fails, the read returns a bus fault.

These checks help to make the OTP fail-safe in contexts where deliberate fault injection is a possibility. For example, the RP2350 bootrom uses guarded reads to check boot configuration flags.

The data returned from a successful guarded read is the same as the data returned by a successful read from the corresponding unguarded alias.

#### ! IMPORTANT

Users relying on OTP data in a Secure context should always perform guarded reads, and it is strongly recommended to use ECC. For rows where ECC is not possible, software should take care to ensure the consistency of data across multiple overlapping reads.

## 13.2. Background: OTP IP Details

The RP2350 OTP subsystem uses the Synopsys NVM OTP IP, which comes in 3 parts:

- Integrated Power Supply (IPS), including:
  - Charge Pump (for programming)
  - Regulator (for reading)
- OTP Macro (SHF, Fuse)
  - $4096 \times 24$  (8 kB with ECC, 16-bit ECC write granularity)
- Access port (AP), providing:

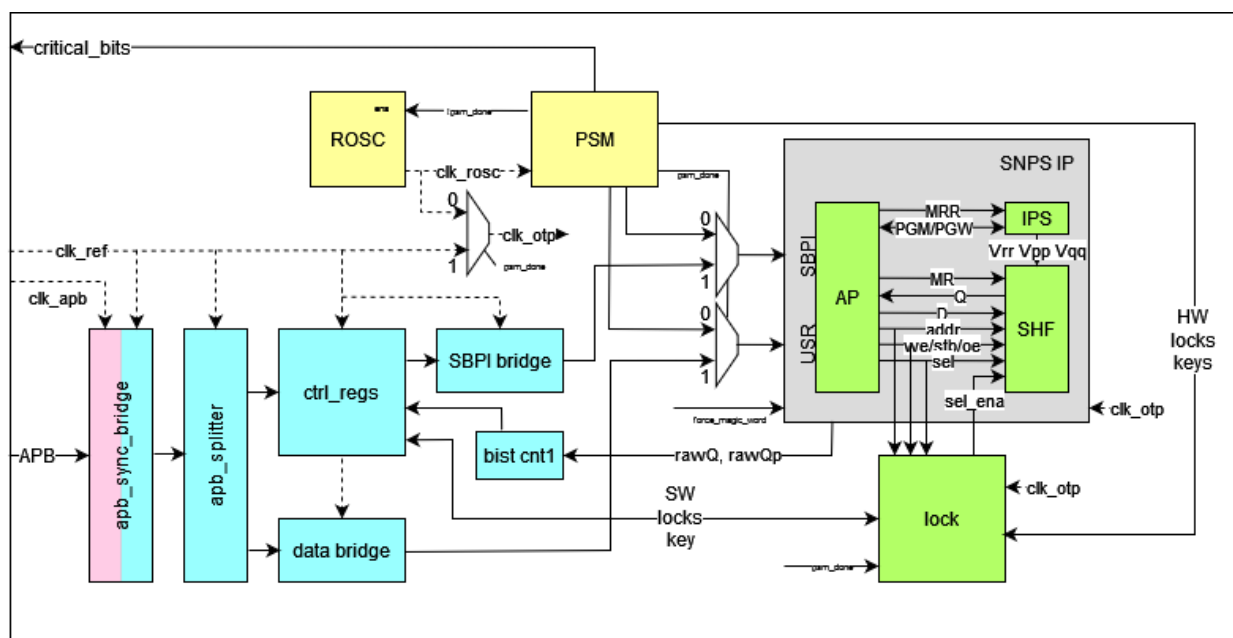
- o Basic read access
- o Programming access
- o ECC and bit redundancy
- o BOOT function, which polls for stable OTP power supply at start-of-day

### 13.3. Background: OTP Hardware Architecture

This diagram shows the integration of the three Synopsys IP components, and the Raspberry Pi hardware added to make this all function in the context of RP2350's system and security architecture. More specifically:

- APB interface(s) to connect to the SoC
- Internal ring oscillator with clock edge randomisation
- Power-up state machine, running off the ring oscillator
- Lock shim, sitting between the SNPS RTL and the memory core (fuse)

Figure 141. OTP architecture



The OTP subsystem clock is initially provided by the OTP boot oscillator (Section 13.3.3) during hardware startup, but switches to `clk_ref` before any software runs on the processors. The frequency of `clk_ref` must not exceed 25 MHz when accessing the OTP.

### 13.3.1. Lock Shim

The lock shim is inserted between the Synopsys AP block and the SHF block, and is used to enforce read/write page locks, based on:

- The OTP address presented on the AP → SHF bus
- The read/write strobe on the AP → SHF bus
- The security attribute of the upstream bus access which caused this SHF access (assumed to be Secure if SBPI is currently enabled via USR.DCTRL)

Because the Synopsys AP performs both reads and writes in the course of programming an OTP row, it is impossible to disable reads to an address without also disabling writes. Three lock states are supported:

- Read/Write
- Read-only
- Inaccessible

The full locking scheme is described in in [Section 13.5](#), but to summarise:

- The lock state of each OTP page is read from OTP at boot time.
- There is a separate copy of the lock state for Secure/Non-secure accesses. The lock shim applies the Secure read permissions to Secure reads, and the Non-secure read permissions to Non-secure reads. There is no such rule for writes, because Non-secure code is not capable of accessing the programming hardware.
- The lock encoding in OTP storage is such that a page can always be locked down to a less permissive state (in the order RW → RO → Inaccessible) but can never return to a more permissive state.
- Software can advance the state of each individual lock at runtime without programming OTP, and this lasts until the OTP PSM is re-run.
- Software locks also obey the lock progression order (RW → RO → Inaccessible) and can not be regressed.

The full locking scheme is described in in [Section 13.5](#).

### 13.3.2. External Interfaces

The OTP integration has one upstream APB interface, which splits internally onto two separate interfaces. This guarantees the hardware only serves a single upstream APB access at a time, with a single PPROT security level.

The first APB interface is the *data* interface (or data bridge) (OTPD). It has the following characteristics:

- Read-only
- Connects to the Synopsys device access port (DAP)
- Data interface reads always return 32 or 24 bits of valid data
- The data interface address is rounded down to a multiple of 32 bits, so that narrow reads return the correct byte lanes
- There is an 8 kB window which supports 32-bit ECC reads
  - Each upstream bus read is split into two OTP accesses, each of which returns 16 bits of error-corrected data from the OTP
- There is an 8 kB window which supports guarded 32-bit ECC reads, and returns a bus error if the guarding read fails.
  - Functions the same as the ECC read window, but reads the Synopsys boot word before accessing the OTP array, and return a bus error if the first read does not match the expected constant
  - Used to increase confidence in software OTP reads in the bootrom
- There is a 16 kB window which supports 24-bit raw reads
  - Each access returns a single raw 24-bit OTP row, bypassing error correction
  - Software must provide its own redundancy (e.g. triple majority vote)
  - Allows bit-mutable data structures, such as boot flags, or thermometer counters

The second APB interface is the *command* interface. This provides two main functions:

- Provides a bridge to the SBPI interface (Synopsys proprietary Serial and Byte-Parallel Interface bus)
  - SBPI connects to the Programmable Master Controller (PMC), with access to the DAP, DATAPATH, charge pump (IPS), and fuse memory (SHF)

- Allows arbitrary OTP operations, including programming
- Only accessible to Secure reads and writes
- Provides control registers for Raspberry Pi hardware
  - Registers have different accessibility according to Secure/Non-secure and read/write
  - Software lock registers are always readable by both security domains

Hardware configuration data read from OTP during the power-up sequence drives system-level control signals, e.g. disabling CoreSight APs. This is described in more detail in (Section 13.3.4).

A single system-level interrupt output (IRQ) generates interrupts for the following sources:

- Secure read failed due to locks
- Non-secure read failed due to locks
- Write failed due to locks
- SBPI FLAG, used by the PMC to signal completion
- Data port access when DCTRL is set error
  - `USR.DCTRL` tells the SNPS AP whether the SBPI bridge or data bridge can access the memory array; this help debugging SW if a data access is attempted whilst the DAP is inaccessible

Any failed access also returns a bus fault (`PSLVERR`). To determine whether an OTP address is accessible, query the lock tables.

Non-secure code cannot access the interrupt status registers.

### 13.3.3. OTP Boot Oscillator

The OTP startup sequence (Section 13.3.4) runs from a local ring oscillator, dedicated to the OTP subsystem. This is separate from the system ring oscillator (the ROSC) which provides the system clock to run the processors during boot.

- The OTP boot oscillator is the only clock used by the OTP power-up state machine
- The OTP boot oscillator dynamically randomises its own frequency controls, to deliberately add jitter to the clock
- The OTP boot oscillator stops when the PSM completes, and does not start again until the OTP resets
- The OTP clock automatically switches to `clk_ref` when the OTP boot oscillator stops

The boot oscillator has a nominal frequency of 12MHz. It provides the clock for reading out hardware configuration from OTP, including the critical flags (Section 13.4) which configure hardware security features such as debug disable and the glitch detectors.

Keeping this oscillator local to the OTP hardware subsystem reduces the power signature of the clock itself, due to the lower switched clock capacitance. Along with the random jitter of the frequency controls, this helps frustrate attempts to recover OTP access keys and debug keys via power signature analysis attacks, or to disable security features by timing fault injection against the OTP clock.

Only the OTP boot oscillator enables the ROSC frequency randomisation feature by default: for later operations using the system ROSC (Section 8.3), you must explicitly enable this feature on that oscillator, by programming the ROSC control registers. The crystal oscillator (XOSC) does not support frequency randomisation.

### 13.3.4. Power-up State Machine

The OTP is the second item in the switched core domain's Power-On State Machine (Section 7.4), after the processor cold reset. OTP does not release its `rst_done`, or enable any debug interface (including the factory test JTAG described in Section 10.10), until the OTP PSM reads out OTP-resident hardware configuration. The `rst_done` output to the system

PSM holds the rest of the system in reset until the OTP PSM completes, so that no software runs until the OTP's contents are known.

The OTP boot sequence runs from a local ring oscillator. This oscillator is dedicated to the OTP subsystem, and is separate from the main system ROSC used by the processors at boot. The sequence is:

1. First, the PSM runs the Synopsys boot instruction. This has the following steps:
  - a. Wait for the power supply to return a 'good' value.
  - b. Read consistency check location until hardware sees the correct value for 16 successive reads.

**NOTE**

Consistency checks use predefined words stored in mask ROM cells with similar analogue properties to OTP cells.

2. Read critical flags (non-ECC): each critical bit is redundant across 8 OTP rows, with three-of-eight vote for each flag.
3. Read hardware access keys via ECC read interface.
4. Read valid bits for hardware access keys, including the debug keys ([Section 3.5.9.2](#))
5. Initialise page lock registers from the lock page via raw read interface.
6. Assert `rst_done` signal to the system power-on state machine
7. The system reset sequence continues, starting with the system ROSC

## 13.4. Critical Flags

Critical flags enable hardware security features which are fundamental to RP2350's secure boot implementation. The OTP power-up state machine reads critical flags very early in the system reset sequence, before any code runs on the processors.

Most critical flags are in the main Boot Configuration page, page 1. These are listed under `CRIT1` in the OTP data listing. The exceptions are the Arm/RISC-V disable flags, which are in the Chip Info page, page 0. This page is made read-only during factory programming, so users can not write to the `CRIT0` flags.

Critical flags define `0` as the unprogrammed value, and `1` as the programmed value. On a blank device, all of the `CRIT1` flags are `0`. The reset value specified below is the value assigned to the internal logic net between the OTP reset being applied and the OTP PSM completing. For example, the reset value of `1` for the debug disable flags implies that debug is not accessible whilst the OTP PSM is running, but may be available afterward, depending on the value read from OTP storage.

- `ARM_DISABLE` (reset: `0`): Force the `ARCHSEL` register to RISC-V, at higher priority than RISC-V disable flag, secure boot enable flag, or default boot architecture flag.
- `RISCV_DISABLE` (reset: `0`): Force the `ARCHSEL` register to Arm, at higher priority than the default boot architecture flag.
- `SECURE_BOOT_ENABLE` (reset: `1`): Enable boot signature checking in bootrom, disable factory JTAG, and force the `ARCHSEL` register to Arm, at higher priority than the default boot architecture flag.
- `SECURE_DEBUG_DISABLE` (reset: `1`): Disable factory JTAG, block Secure accesses from Mem-APs, and block halt requests to Secure processors.
  - Prevents secure AP accesses by masking their `ap_secure_en` signals.
  - Prevents secure processor halting by masking the Cortex-M33's `SPIDEN` and `NSPIDEN` signals.
  - Secure debug can be re-enabled by a Secure register in the OTP block.
  - Re-enable of Secure debug can be disabled by a Secure write-1-only lock register, also in the OTP block.

- **DEBUG\_DISABLE** (reset: 1): Completely disable the Mem-APs, in addition to disabling everything disabled by the secure debug disable flag.
- **BOOT\_ARCH** (reset: 0): set the reset value of the **ARCHSEL** register (0 → Arm, 1 → RISC-V) if it has not been forced by other critical flags.
  - Not critical, but hardware-read.
- **GLITCH\_DETECTOR\_ENABLE** (reset: 0): pass an enable signal to the glitch detectors so that they can be armed before any software runs.
- **GLITCH\_DETECTOR\_SENS**(reset: 0): configure the initial sensitivity of the glitch detector circuits.

Critical flags are encoded with a three-of-eight vote across eight consecutive OTP rows. Each flag is redundantly programmed to the same bit position in eight consecutive rows. Hardware considers the flag to be set if the bit reads as 1 in at least three of these eight rows. The flag is considered clear if no more than two bits are observed to be set.

JTAG disable is ignored only if the customer RMA flag ([Section 13.7](#)) is set.

For further discussion of the effects of the critical flags, see:

- [Section 3.5.9.1](#) for the effects of the debug disable flags
- [Section 3.9](#) for the effects of the Arm/RISC-V architecture select flags
- [Section 10.9](#) for the effects of the glitch detector configuration flags
- [Section 10.1.1](#) for discussion of the bootrom secure boot support enabled by the **SECURE\_BOOT\_ENABLE** flag

## 13.5. Page Locks

The OTP protection hardware logically segments OTP into 64 *pages* (0 through 63), each 128 bytes in size, or equivalently 64 OTP rows.

Each page has a set of lock registers which determine read and write access for that page from Secure and Non-secure code. The lock registers are preloaded from OTP at reset, and can then be advanced (i.e. made less permissive) by software. Lock registers themselves are always world-readable.

Pages 61 through 63 are not so neatly described by a single set of lock registers. These pages store lock initialisation metadata. For more details, see [Section 13.5.4](#). This section describes the more common case of a page protected by a set of page locks.

### 13.5.1. Lock Progression

Due to hardware constraints ([Section 13.3.1](#)), read and write restrictions are not orthogonal: it's impossible to disallow reads to an address without also disallowing writes. So, the progression of locking for a given page is:

0. Read/Write
1. Read-only
2. Inaccessible

Lock state *only increases*. This is enforced in two ways:

- Due to the nature of OTP and the choice of encoding, you cannot lower the OTP values preloaded to the lock registers during boot.
- The lock registers ignore writes of lower-than-current values.

Secure and Non-secure use separate lock values, which can advance independently of one another. There is no hardware distinction between Non-secure Read/Write and Non-secure Read-only, since Non-secure can not directly write to the OTP anyway. It is still worth encoding, because Secure software performing a write on Non-secure software's behalf can check and enforce the Non-secure write lock.

You can reprogram bits from any state to any higher state. Locks use a 2-bit thermometer code: the initial all-zeroes state is read-write, and locks are advanced by programming first bit 0, then bit 1.

Lock bits in OTP are triple-redundant with a majority vote. They can't be ECC-protected, because they may be mutated bit-by-bit over multiple programming operations.

The OTP-resident lock bits are write-protected by their own Secure lock level. The lock pages are always world-readable.

The Secure lock registers can be advanced by Secure code, and are world-readable.

The Non-secure lock registers can be advanced by Secure or Non-secure code, and are world-readable.

### 13.5.2. OTP Access Keys

Page 61 contains 128-bit keys. Each key has a **valid bit**: when set, the key becomes completely inaccessible to software. The keys are always read out into hidden registers by hardware during startup so that hardware can perform key comparisons without exposing the keys to software.

Pages can require specific keys for some page permissions. To unlock the page, the user writes their key to a write-only register in the OTP block. The page remains unlocked for as long as the correct key is present in this register. To re-lock the page, erase the active key by writing zeroes to the key register.

The per-page lock config specifies the following:

- a **read key** index 1-7, or 0 if there is no read key
- a **write key** index 1-7 or 0 if there is no write key
- the **no-key state**: either Read-only or Inaccessible, state of the page when no registered key has been entered by software into the key register

The no-key state is encoded as follows:

- 0 for Read-only (lock level 1)
- 1 for Inaccessible (lock level 2).

#### TIP

Key index 7 does not exist in the configuration. If you specify key index 7, it is guaranteed to never match.

The hardware determines the **key lock level** by comparing the entered key to the key config of the current page, as follows:

1. If no keys are registered, the key lock level is 0
2. Else if keys are registered and no matching key is entered, the key lock level is 2 or 1 depending on the "no-key state" config
3. Else if a write key is registered and present, the key lock level is 0
4. Else if a read key is registered and present, the key lock level is 1

Hardware compares the key lock level to the page's lock level for the current security domain (Secure/Non-secure) and takes **whichever is higher**. For example, if a page has been made Non-secure read-only, there is nothing a key can do to make it Non-secure writable.

There are six 128-bit access keys stored in the OTP. Keys 5 and 6 also function as the Secure debug access key and Non-secure debug access key, respectively. See [Section 3.5.9.2](#) for information on how the debug keys affect external debug access.

You might use OTP access keys if a bootloader contains OTP configuration that needs to be Secure-writable *only to the board owner*, not to general Secure software on the device.

### 13.5.3. Lock Encoding in OTP

Page locks are encoded as a 16-bit value. This value is stored as a pair of triple-redundant bytes, each byte occupying a 24-bit OTP row.

The lock halfword is encoded as follows:

| Bits  | Purpose                                                                                          |
|-------|--------------------------------------------------------------------------------------------------|
| 2:0   | Write key index, or 0 if no write key                                                            |
| 5:3   | Read key index, or 0 if no read key                                                              |
| 6     | No-key state, 0=Read-only 1=Inaccessible                                                         |
| 7     | Reserved                                                                                         |
| 9:8   | Secure lock state (thermometer code 0 → 2)                                                       |
| 11:10 | Non-secure lock state (thermometer code 0 → 2)                                                   |
| 13:12 | PicoBoot lock state (thermometer code 0 → 2) or software-defined use if PicoBoot OTP is disabled |
| 15:12 | Reserved                                                                                         |

### 13.5.4. Special Pages

The following pages require special case handling in their lock checks:

- The lock word region itself (pages 62 and 63)
  - Lock words are always world-readable
  - Lock words are writable by Secure code if the lock word itself permits Secure writes
  - Consequently, lock words 62 and 63 are considered "spare", since they do not protect pages 62 and 63; the page 63 lock word is repurposed for the RMA flag
- The hardware access key page (page 61)
  - Contains OTP access keys and debug access keys
  - Each key also has a valid bit (rbit)
  - Page 61 (key page) has all of the usual protections from the page 61 lock word
  - If a key's valid bit is set, that key is inaccessible; the converse is not necessarily true

Page 0, known as the **chip info page**, is *not* a special page. Raspberry Pi sets page 0 to read-only during factory test, after writing chip identification and calibration values.

### 13.5.5. Permissions of Blank Devices

Each RP2350 device has some information programmed during manufacturing test. At this time, a small number of hard page lock bits are also programmed:

- Page 0, which contains chip information, is read-only for all accesses.
- Pages 1 and 2, which contain boot config and boot key fingerprints, are read-only for Non-secure access, read-write for Secure access, and read-write for bootloader access.
- Page 63, which contains the RMA flag, is read-only for Non-secure and bootloader access, and read-write for Secure access.

This minimal set of default permissions on blank devices avoids certain classes of security model violation, like Non-

secure code being able to brick the chip by overwriting the boot key fingerprints with invalid data. In this context, the term *blank device* refers to a device that has gone through manufacturing test programming, but has not had any other OTP bits programmed by the user.

You can add additional soft or hard locks to these default permissions, with the exception of page 0. Page 0 cannot be hard-locked, since the secure read-only permission prevents a user from altering its lock word.

Lock words 2 through 61, covering all pages with user-defined contents, are left unprogrammed. On a blank device, these pages are fully accessible from all domains. Before launching any Non-secure application, you should apply at least a soft read-only lock to all pages that are not explicitly allocated for Non-secure use. To do this, write to [SW\\_LOCK2](#) through [SW\\_LOCK61](#). For devices that you don't expect to RMA, such as those that have passed board-level manufacturing tests, you should lock secure writes to the RMA flag.

## 13.6. Error Correction Code (ECC)

ECC-protected rows store data in the following structure, accessible through a raw alias:

- Bits [23:22](#): bit repair by polarity (BRP) flag
- Bits [21:16](#): modified Hamming ECC code
- Bits [15:0](#) (the 16 LSBs): data

RP2350 stores the following error correction data in the 8 MSBs of each 24-bit row:

- a 6-bit modified Hamming code ECC, providing single-error-correct and double-error-detect capabilities
- 2 bits of bit repair by polarity (BRP), which supports inverting the entire row at programming time to repair a single set bit that should be clear

Writes first encode ECC, then BRP. Reads first decode BRP, then ECC. When reading through an ECC data alias ([Section 13.1](#)), hardware performs correction transparently. ECC programming operations (writes) automatically generate ECC bits when you use the bootrom [otp\\_access](#) API ([Section 5.4.8.21](#)).

ECC is not suitable for data that mutates one bit at a time, since the ECC value is derived from the entire 16-bit data value. When storing data without ECC, use another form of redundancy, such as 3-way majority vote.

### 13.6.1. Bit repair by polarity (BRP)

*Bit repair by polarity (BRP)* compensates for a single bit present at time of programming.

When programming a row, hardware or software first calculates a 24-bit target value consisting of:

- Two zeroes in bits [23:22](#)
- 6-bit Hamming ECC in bits [21:16](#)
- 16-bit data value in bits [15:0](#)

Before programming, an OTP row should contain all zeros. However, sometimes OTP rows contain a single bit that is already set to **1**, either due to manufacturing flaws or previous programming. If a bit is already set (**1**) in an OTP row before programming, BRP checks the status of the corresponding bit in the target value. BRP compensates for this single set bit in one of two ways, depending on the corresponding value in the target value:

- If the bit is clear (**0**), BRP inverts the target row and writes two ones in bits [23:22](#).
- If the bit is set (**1**), BRP does not invert the target row, leaving two zeroes in bits [23:22](#).

When you read an OTP value through an ECC alias ([Section 13.1](#)), BRP checks for two ones in bits [23:22](#). When both bits 23 and 22 are set, BRP inverts the entire row before passing it to the modified Hamming code stage.

BRP makes it possible to store any 22-bit value in a row that initially has at most one bit set, preserving the correction margin of the modified Hamming code. During manufacturing test, hardware scans the entire OTP array to ensure no

rows contain more than one pre-set bit.

### 13.6.2. Modified Hamming ECC

ECC generates six **parity bits** based on the data value stored in bits **15:0** of an OTP row. When programming a row, ECC generates those six parity bits and includes them in the target value as bits **21:16**. This code consists of:

- A 5-bit Hamming code that identifies single-bit errors
- An even parity bit which allows two-bit errors to be detected in the Hamming code and the original 16-bit data

When you read an OTP value through an ECC alias ([Section 13.1](#)), ECC recalculates the six parity bits based on the value read from the OTP row. Then, ECC XORs the original six parity bits with the newly-calculated parity bits. This generates 6 new bits:

- the 5 LSBs are the *syndrome*, a unique bit pattern that corresponds to each possible bit flip in the data value
- the MSB distinguishes between odd and even numbers of bit flips

If all 6 bits in this value are zero, ECC did not detect an error. If the MSB is 1, the syndrome should indicate a single-bit error. ECC flips the corresponding data bit to recover from the error. If the MSB is 0, but the syndrome contains a value other than 0, the ECC detected an unrecoverable multi-bit error.

You can calculate 5-bit Hamming codes and parity bits with the following C code (adapted from the RP2350 bootrom source):

```
uint32_t even_parity(uint32_t input) {
    uint32_t rc = 0;
    while (input) {
        rc ^= input & 1;
        input >>= 1;
    }
    return rc;
}

const uint32_t otp_ecc_parity_table[6] = {
    0b00000010101101011011,
    0b000000011011001101101,
    0b0000001100011110001110,
    0b000000000001111110000,
    0b000000111110000000000,
    0b01111111111111111111
};

uint32_t s_otp_calculate_ecc(uint16_t x) {
    uint32_t p = x;
    for (uint i = 0; i < 6; ++i) {
        p |= even_parity(p & otp_ecc_parity_table[i]) << (16 + i);
    }
    return p;
}
```

## 13.7. Device Decommissioning (RMA)

Decommissioning refers to destroying a device's sensitive contents and restoring some test or debug functionality when a device reaches the end of its security lifecycle. The OTP hardware can't actually destroy user data without circumventing write protection in some way. Instead, decommissioning is implemented with the **RMA flag**, which modifies devices in the following ways:

- re-enables factory test JTAG which is otherwise disabled by the secure boot critical flag
- makes pages 3 through 61 inaccessible

The RMA flag doesn't change permissions for page 0 (manufacturing data), pages 1 and 2 (boot configuration), page 61 (OTP access keys), or pages 62 and 63 (locks).

The RMA flag is encoded in a spare bit of the page 63 lock word. This lock word would otherwise be unused, since page 63 is one of the lock pages; consequently, it is not protected by a lock word. Instead, each lock word protects itself.

Like all other lock words, the page 63 lock word is protected by its own locks, which means it can be hard- and soft-locked to prevent the RMA flag being set. Locking the RMA flag makes it impossible to re-enable the factory JTAG interface if any of [CRIT1.SECURE\\_BOOT\\_ENABLE](#), [CRIT1.DEBUG\\_DISABLE](#) or [CRIT1.SECURE\\_DEBUG\\_DISABLE](#) is set. This makes it impossible for Raspberry Pi to re-test such devices if they are returned for fault analysis.

### ! IMPORTANT

Setting the RMA flag does not destroy OTP contents, it merely renders it inaccessible. The design intent is for this to be irreversible, but hardware is never perfect. This is something the user's threat model must account for when programming the RMA flag on devices with sensitive OTP contents – for example, by personalising per-device OTP secrets to avoid class breaks if an attacker is able to retrieve the keys.

## 13.8. List of Registers

The OTP control registers start at a base address of [0x40120000](#) (defined as [OTP\\_BASE](#) in the SDK).

Table 1331. List of OTP registers

| Offset | Name                      | Info                                |
|--------|---------------------------|-------------------------------------|
| 0x000  | <a href="#">SW_LOCK0</a>  | Software lock register for page 0.  |
| 0x004  | <a href="#">SW_LOCK1</a>  | Software lock register for page 1.  |
| 0x008  | <a href="#">SW_LOCK2</a>  | Software lock register for page 2.  |
| 0x00c  | <a href="#">SW_LOCK3</a>  | Software lock register for page 3.  |
| 0x010  | <a href="#">SW_LOCK4</a>  | Software lock register for page 4.  |
| 0x014  | <a href="#">SW_LOCK5</a>  | Software lock register for page 5.  |
| 0x018  | <a href="#">SW_LOCK6</a>  | Software lock register for page 6.  |
| 0x01c  | <a href="#">SW_LOCK7</a>  | Software lock register for page 7.  |
| 0x020  | <a href="#">SW_LOCK8</a>  | Software lock register for page 8.  |
| 0x024  | <a href="#">SW_LOCK9</a>  | Software lock register for page 9.  |
| 0x028  | <a href="#">SW_LOCK10</a> | Software lock register for page 10. |
| 0x02c  | <a href="#">SW_LOCK11</a> | Software lock register for page 11. |
| 0x030  | <a href="#">SW_LOCK12</a> | Software lock register for page 12. |
| 0x034  | <a href="#">SW_LOCK13</a> | Software lock register for page 13. |
| 0x038  | <a href="#">SW_LOCK14</a> | Software lock register for page 14. |
| 0x03c  | <a href="#">SW_LOCK15</a> | Software lock register for page 15. |
| 0x040  | <a href="#">SW_LOCK16</a> | Software lock register for page 16. |
| 0x044  | <a href="#">SW_LOCK17</a> | Software lock register for page 17. |
| 0x048  | <a href="#">SW_LOCK18</a> | Software lock register for page 18. |

| Offset | Name                      | Info                                |
|--------|---------------------------|-------------------------------------|
| 0x04c  | <a href="#">SW_LOCK19</a> | Software lock register for page 19. |
| 0x050  | <a href="#">SW_LOCK20</a> | Software lock register for page 20. |
| 0x054  | <a href="#">SW_LOCK21</a> | Software lock register for page 21. |
| 0x058  | <a href="#">SW_LOCK22</a> | Software lock register for page 22. |
| 0x05c  | <a href="#">SW_LOCK23</a> | Software lock register for page 23. |
| 0x060  | <a href="#">SW_LOCK24</a> | Software lock register for page 24. |
| 0x064  | <a href="#">SW_LOCK25</a> | Software lock register for page 25. |
| 0x068  | <a href="#">SW_LOCK26</a> | Software lock register for page 26. |
| 0x06c  | <a href="#">SW_LOCK27</a> | Software lock register for page 27. |
| 0x070  | <a href="#">SW_LOCK28</a> | Software lock register for page 28. |
| 0x074  | <a href="#">SW_LOCK29</a> | Software lock register for page 29. |
| 0x078  | <a href="#">SW_LOCK30</a> | Software lock register for page 30. |
| 0x07c  | <a href="#">SW_LOCK31</a> | Software lock register for page 31. |
| 0x080  | <a href="#">SW_LOCK32</a> | Software lock register for page 32. |
| 0x084  | <a href="#">SW_LOCK33</a> | Software lock register for page 33. |
| 0x088  | <a href="#">SW_LOCK34</a> | Software lock register for page 34. |
| 0x08c  | <a href="#">SW_LOCK35</a> | Software lock register for page 35. |
| 0x090  | <a href="#">SW_LOCK36</a> | Software lock register for page 36. |
| 0x094  | <a href="#">SW_LOCK37</a> | Software lock register for page 37. |
| 0x098  | <a href="#">SW_LOCK38</a> | Software lock register for page 38. |
| 0x09c  | <a href="#">SW_LOCK39</a> | Software lock register for page 39. |
| 0x0a0  | <a href="#">SW_LOCK40</a> | Software lock register for page 40. |
| 0x0a4  | <a href="#">SW_LOCK41</a> | Software lock register for page 41. |
| 0x0a8  | <a href="#">SW_LOCK42</a> | Software lock register for page 42. |
| 0x0ac  | <a href="#">SW_LOCK43</a> | Software lock register for page 43. |
| 0x0b0  | <a href="#">SW_LOCK44</a> | Software lock register for page 44. |
| 0x0b4  | <a href="#">SW_LOCK45</a> | Software lock register for page 45. |
| 0x0b8  | <a href="#">SW_LOCK46</a> | Software lock register for page 46. |
| 0x0bc  | <a href="#">SW_LOCK47</a> | Software lock register for page 47. |
| 0x0c0  | <a href="#">SW_LOCK48</a> | Software lock register for page 48. |
| 0x0c4  | <a href="#">SW_LOCK49</a> | Software lock register for page 49. |
| 0x0c8  | <a href="#">SW_LOCK50</a> | Software lock register for page 50. |
| 0x0cc  | <a href="#">SW_LOCK51</a> | Software lock register for page 51. |
| 0x0d0  | <a href="#">SW_LOCK52</a> | Software lock register for page 52. |
| 0x0d4  | <a href="#">SW_LOCK53</a> | Software lock register for page 53. |
| 0x0d8  | <a href="#">SW_LOCK54</a> | Software lock register for page 54. |

| Offset | Name                         | Info                                                                                                                                                                                                                                                                              |
|--------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x0dc  | <a href="#">SW_LOCK55</a>    | Software lock register for page 55.                                                                                                                                                                                                                                               |
| 0x0e0  | <a href="#">SW_LOCK56</a>    | Software lock register for page 56.                                                                                                                                                                                                                                               |
| 0x0e4  | <a href="#">SW_LOCK57</a>    | Software lock register for page 57.                                                                                                                                                                                                                                               |
| 0x0e8  | <a href="#">SW_LOCK58</a>    | Software lock register for page 58.                                                                                                                                                                                                                                               |
| 0x0ec  | <a href="#">SW_LOCK59</a>    | Software lock register for page 59.                                                                                                                                                                                                                                               |
| 0x0f0  | <a href="#">SW_LOCK60</a>    | Software lock register for page 60.                                                                                                                                                                                                                                               |
| 0x0f4  | <a href="#">SW_LOCK61</a>    | Software lock register for page 61.                                                                                                                                                                                                                                               |
| 0x0f8  | <a href="#">SW_LOCK62</a>    | Software lock register for page 62.                                                                                                                                                                                                                                               |
| 0x0fc  | <a href="#">SW_LOCK63</a>    | Software lock register for page 63.                                                                                                                                                                                                                                               |
| 0x100  | <a href="#">SBPI_INSTR</a>   | Dispatch instructions to the SBPI interface, used for programming the OTP fuses.                                                                                                                                                                                                  |
| 0x104  | <a href="#">SBPI_WDATA_0</a> | SBPI write payload bytes 3..0                                                                                                                                                                                                                                                     |
| 0x108  | <a href="#">SBPI_WDATA_1</a> | SBPI write payload bytes 7..4                                                                                                                                                                                                                                                     |
| 0x10c  | <a href="#">SBPI_WDATA_2</a> | SBPI write payload bytes 11..8                                                                                                                                                                                                                                                    |
| 0x110  | <a href="#">SBPI_WDATA_3</a> | SBPI write payload bytes 15..12                                                                                                                                                                                                                                                   |
| 0x114  | <a href="#">SBPI_RDATA_0</a> | Read payload bytes 3..0. Once read, the data in the register will automatically clear to 0.                                                                                                                                                                                       |
| 0x118  | <a href="#">SBPI_RDATA_1</a> | Read payload bytes 7..4. Once read, the data in the register will automatically clear to 0.                                                                                                                                                                                       |
| 0x11c  | <a href="#">SBPI_RDATA_2</a> | Read payload bytes 11..8. Once read, the data in the register will automatically clear to 0.                                                                                                                                                                                      |
| 0x120  | <a href="#">SBPI_RDATA_3</a> | Read payload bytes 15..12. Once read, the data in the register will automatically clear to 0.                                                                                                                                                                                     |
| 0x124  | <a href="#">SBPI_STATUS</a>  |                                                                                                                                                                                                                                                                                   |
| 0x128  | <a href="#">USR</a>          | Controls for APB data read interface (USER interface)                                                                                                                                                                                                                             |
| 0x12c  | <a href="#">DBG</a>          | Debug for OTP power-on state machine                                                                                                                                                                                                                                              |
| 0x134  | <a href="#">BIST</a>         | During BIST, count address locations that have at least one leaky bit                                                                                                                                                                                                             |
| 0x138  | <a href="#">CRT_KEY_W0</a>   | Word 0 (bits 31..0) of the key. Write only, read returns 0x0                                                                                                                                                                                                                      |
| 0x13c  | <a href="#">CRT_KEY_W1</a>   | Word 1 (bits 63..32) of the key. Write only, read returns 0x0                                                                                                                                                                                                                     |
| 0x140  | <a href="#">CRT_KEY_W2</a>   | Word 2 (bits 95..64) of the key. Write only, read returns 0x0                                                                                                                                                                                                                     |
| 0x144  | <a href="#">CRT_KEY_W3</a>   | Word 3 (bits 127..96) of the key. Write only, read returns 0x0                                                                                                                                                                                                                    |
| 0x148  | <a href="#">CRITICAL</a>     | Quickly check values of critical flags read during boot up                                                                                                                                                                                                                        |
| 0x14c  | <a href="#">KEY_VALID</a>    | Which keys were valid (enrolled) at boot time                                                                                                                                                                                                                                     |
| 0x150  | <a href="#">DEBUGEN</a>      | Enable a debug feature that has been disabled. Debug features are disabled if one of the relevant critical boot flags is set in OTP (DEBUG_DISABLE or SECURE_DEBUG_DISABLE), OR if a debug key is marked valid in OTP, and the matching key value has not been supplied over SWD. |

| Offset | Name                           | Info                                                                                                                                                                                                                      |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x154  | <a href="#">DEBUGEN_LOCK</a>   | Write 1s to lock corresponding bits in DEBUGEN. This register is reset by the processor cold reset.                                                                                                                       |
| 0x158  | <a href="#">ARCHSEL</a>        | Architecture select (Arm/RISC-V), applied on next processor reset. The default and allowable values of this register are constrained by the critical boot flags.                                                          |
| 0x15c  | <a href="#">ARCHSEL_STATUS</a> | Get the current architecture select state of each core. Cores sample the current value of the ARCHSEL register when their warm reset is released, at which point the corresponding bit in this register will also update. |
| 0x160  | <a href="#">BOOTDIS</a>        | Tell the bootrom to ignore scratch register boot vectors (both power manager and watchdog) on the next power up.                                                                                                          |
| 0x164  | <a href="#">INTR</a>           | Raw Interrupts                                                                                                                                                                                                            |
| 0x168  | <a href="#">INTE</a>           | Interrupt Enable                                                                                                                                                                                                          |
| 0x16c  | <a href="#">INTF</a>           | Interrupt Force                                                                                                                                                                                                           |
| 0x170  | <a href="#">INTS</a>           | Interrupt status after masking & forcing                                                                                                                                                                                  |

## OTP: SW\_LOCK0, SW\_LOCK1, ..., SW\_LOCK62, SW\_LOCK63 Registers

**Offsets:** 0x000, 0x004, ..., 0x0f8, 0x0fc

### Description

Software lock register for page *N*.

Locks are initialised from the OTP lock pages at reset. This register can be written to further advance the lock state of each page (until next reset), and read to check the current lock state of a page.

Table 1332.  
SW\_LOCK0,  
SW\_LOCK1, ...,  
SW\_LOCK62,  
SW\_LOCK63 Registers

| Bits | Description                                                                                                         | Type | Reset |
|------|---------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:4 | Reserved.                                                                                                           | -    | -     |
| 3:2  | <b>NSEC:</b> Non-secure lock status. Writes are OR'd with the current value.                                        | RW   | -     |
|      | Enumerated values:                                                                                                  |      |       |
|      | 0x0 → READ_WRITE                                                                                                    |      |       |
|      | 0x1 → READ_ONLY                                                                                                     |      |       |
|      | 0x3 → INACCESSIBLE                                                                                                  |      |       |
| 1:0  | <b>SEC:</b> Secure lock status. Writes are OR'd with the current value. This field is read-only to Non-secure code. | RW   | -     |
|      | Enumerated values:                                                                                                  |      |       |
|      | 0x0 → READ_WRITE                                                                                                    |      |       |
|      | 0x1 → READ_ONLY                                                                                                     |      |       |
|      | 0x3 → INACCESSIBLE                                                                                                  |      |       |

## OTP: SBPI\_INSTR Register

**Offset:** 0x100

### Description

Dispatch instructions to the SBPI interface, used for programming the OTP fuses.

Table 1333.  
SBPL\_INSTR Register

| Bits  | Description                                                                     | Type | Reset |
|-------|---------------------------------------------------------------------------------|------|-------|
| 31    | Reserved.                                                                       | -    | -     |
| 30    | <b>EXEC</b> : Execute instruction                                               | SC   | 0x0   |
| 29    | <b>IS_WR</b> : Payload type is write                                            | RW   | 0x0   |
| 28    | <b>HAS_PAYLOAD</b> : Instruction has payload (data to be written or to be read) | RW   | 0x0   |
| 27:24 | <b>PAYLOAD_SIZE_M1</b> : Instruction payload size in bytes minus 1              | RW   | 0x0   |
| 23:16 | <b>TARGET</b> : Instruction target, it can be PMC (0x3a) or DAP (0x02)          | RW   | 0x00  |
| 15:8  | <b>CMD</b>                                                                      | RW   | 0x00  |
| 7:0   | <b>SHORT_WDATA</b> : wdata to be used only when payload_size_m1=0               | RW   | 0x00  |

## OTP: SBPI\_WDATA\_0 Register

Offset: 0x104

Table 1334.  
SBPL\_WDATA\_0  
Register

| Bits | Description                   | Type | Reset      |
|------|-------------------------------|------|------------|
| 31:0 | SBPI write payload bytes 3..0 | RW   | 0x00000000 |

## OTP: SBPI\_WDATA\_1 Register

Offset: 0x108

Table 1335.  
SBPL\_WDATA\_1  
Register

| Bits | Description                   | Type | Reset      |
|------|-------------------------------|------|------------|
| 31:0 | SBPI write payload bytes 7..4 | RW   | 0x00000000 |

## OTP: SBPI\_WDATA\_2 Register

Offset: 0x10c

Table 1336.  
SBPL\_WDATA\_2  
Register

| Bits | Description                    | Type | Reset      |
|------|--------------------------------|------|------------|
| 31:0 | SBPI write payload bytes 11..8 | RW   | 0x00000000 |

## OTP: SBPI\_WDATA\_3 Register

Offset: 0x110

Table 1337.  
SBPL\_WDATA\_3  
Register

| Bits | Description                     | Type | Reset      |
|------|---------------------------------|------|------------|
| 31:0 | SBPI write payload bytes 15..12 | RW   | 0x00000000 |

## OTP: SBPI\_RDATA\_0 Register

Offset: 0x114

Table 1338.  
SBPL\_RDATA\_0  
Register

| Bits | Description                                                                                 | Type | Reset      |
|------|---------------------------------------------------------------------------------------------|------|------------|
| 31:0 | Read payload bytes 3..0. Once read, the data in the register will automatically clear to 0. | RO   | 0x00000000 |

## OTP: SBPI\_RDATA\_1 Register

Offset: 0x118

Table 1339.  
SBPL\_RDATA\_1  
Register

| Bits | Description                                                                                 | Type | Reset      |
|------|---------------------------------------------------------------------------------------------|------|------------|
| 31:0 | Read payload bytes 7..4. Once read, the data in the register will automatically clear to 0. | RO   | 0x00000000 |

## OTP: SBPI\_RDATA\_2 Register

Offset: 0x11c

Table 1340.  
SBPL\_RDATA\_2  
Register

| Bits | Description                                                                                  | Type | Reset      |
|------|----------------------------------------------------------------------------------------------|------|------------|
| 31:0 | Read payload bytes 11..8. Once read, the data in the register will automatically clear to 0. | RO   | 0x00000000 |

## OTP: SBPI\_RDATA\_3 Register

Offset: 0x120

Table 1341.  
SBPL\_RDATA\_3  
Register

| Bits | Description                                                                                   | Type | Reset      |
|------|-----------------------------------------------------------------------------------------------|------|------------|
| 31:0 | Read payload bytes 15..12. Once read, the data in the register will automatically clear to 0. | RO   | 0x00000000 |

## OTP: SBPI\_STATUS Register

Offset: 0x124

Table 1342.  
SBPL\_STATUS Register

| Bits  | Description                                                                                     | Type | Reset |
|-------|-------------------------------------------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                                                       | -    | -     |
| 23:16 | <b>MISO</b> : SBPI MISO (master in - slave out): response from SBPI                             | RO   | -     |
| 15:13 | Reserved.                                                                                       | -    | -     |
| 12    | <b>FLAG</b> : SBPI flag                                                                         | RO   | -     |
| 11:9  | Reserved.                                                                                       | -    | -     |
| 8     | <b>INSTR_MISS</b> : Last instruction missed (dropped), as the previous has not finished running | WC   | 0x0   |
| 7:5   | Reserved.                                                                                       | -    | -     |
| 4     | <b>INSTR_DONE</b> : Last instruction done                                                       | WC   | 0x0   |
| 3:1   | Reserved.                                                                                       | -    | -     |
| 0     | <b>RDATA_VLD</b> : Read command has returned data                                               | WC   | 0x0   |

## OTP: USR Register

Offset: 0x128

### Description

Controls for APB data read interface (USER interface)

Table 1343. USR  
Register

| Bits | Description                                                                                | Type | Reset |
|------|--------------------------------------------------------------------------------------------|------|-------|
| 31:5 | Reserved.                                                                                  | -    | -     |
| 4    | <b>PD</b> : Power-down; 1 disables current reference. Must be 0 to read data from the OTP. | RW   | 0x0   |
| 3:1  | Reserved.                                                                                  | -    | -     |

| Bits | Description                                                                                                                                                                                                                                                                                                                       | Type | Reset |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 0    | <b>DCTRL</b> : 1 enables USER interface; 0 disables USER interface (enables SBPI).<br><br>This bit must be cleared before performing any SBPI access, such as when programming the OTP. The APB data read interface (USER interface) will be inaccessible during this time, and will return a bus error if any read is attempted. | RW   | 0x1   |

## OTP: DBG Register

**Offset:** 0x12c

### Description

Debug for OTP power-on state machine

Table 1344. DBG Register

| Bits  | Description                                                   | Type | Reset |
|-------|---------------------------------------------------------------|------|-------|
| 31:13 | Reserved.                                                     | -    | -     |
| 12    | <b>CUSTOMER_RMA_FLAG</b> : The chip is in RMA mode            | RO   | -     |
| 11:8  | Reserved.                                                     | -    | -     |
| 7:4   | <b>PSM_STATE</b> : Monitor the PSM FSM's state                | RO   | -     |
| 3     | <b>ROSC_UP</b> : Ring oscillator is up and running            | RO   | -     |
| 2     | <b>ROSC_UP_SEEN</b> : Ring oscillator was seen up and running | WC   | 0x0   |
| 1     | <b>BOOT_DONE</b> : PSM boot done status flag                  | RO   | -     |
| 0     | <b>PSM_DONE</b> : PSM done status flag                        | RO   | -     |

## OTP: BIST Register

**Offset:** 0x134

### Description

During BIST, count address locations that have at least one leaky bit

Table 1345. BIST Register

| Bits  | Description                                                                                                                                        | Type | Reset  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|
| 31    | Reserved.                                                                                                                                          | -    | -      |
| 30    | <b>CNT_FAIL</b> : Flag if the count of address locations with at least one leaky bit exceeds cnt_max                                               | RO   | -      |
| 29    | <b>CNT_CLR</b> : Clear counter before use                                                                                                          | SC   | 0x0    |
| 28    | <b>CNT_ENA</b> : Enable the counter before the BIST function is initiated                                                                          | RW   | 0x0    |
| 27:16 | <b>CNT_MAX</b> : The cnt_fail flag will be set if the number of leaky locations exceeds this number                                                | RW   | 0xffff |
| 15:13 | Reserved.                                                                                                                                          | -    | -      |
| 12:0  | <b>CNT</b> : Number of locations that have at least one leaky bit. Note: This count is true only if the BIST was initiated without the fix option. | RO   | -      |

## OTP: CRT\_KEY\_W0 Register

**Offset:** 0x138

Table 1346.  
CRT\_KEY\_W0 Register

| Bits | Description                                                  | Type | Reset      |
|------|--------------------------------------------------------------|------|------------|
| 31:0 | Word 0 (bits 31..0) of the key. Write only, read returns 0x0 | WO   | 0x00000000 |

## OTP: CRT\_KEY\_W1 Register

Offset: 0x13c

Table 1347.  
CRT\_KEY\_W1 Register

| Bits | Description                                                   | Type | Reset      |
|------|---------------------------------------------------------------|------|------------|
| 31:0 | Word 1 (bits 63..32) of the key. Write only, read returns 0x0 | WO   | 0x00000000 |

## OTP: CRT\_KEY\_W2 Register

Offset: 0x140

Table 1348.  
CRT\_KEY\_W2 Register

| Bits | Description                                                   | Type | Reset      |
|------|---------------------------------------------------------------|------|------------|
| 31:0 | Word 2 (bits 95..64) of the key. Write only, read returns 0x0 | WO   | 0x00000000 |

## OTP: CRT\_KEY\_W3 Register

Offset: 0x144

Table 1349.  
CRT\_KEY\_W3 Register

| Bits | Description                                                    | Type | Reset      |
|------|----------------------------------------------------------------|------|------------|
| 31:0 | Word 3 (bits 127..96) of the key. Write only, read returns 0x0 | WO   | 0x00000000 |

## OTP: CRITICAL Register

Offset: 0x148

### Description

Quickly check values of critical flags read during boot up

Table 1350. CRITICAL Register

| Bits  | Description            | Type | Reset |
|-------|------------------------|------|-------|
| 31:18 | Reserved.              | -    | -     |
| 17    | RISCV_DISABLE          | RO   | 0x0   |
| 16    | ARM_DISABLE            | RO   | 0x0   |
| 15:7  | Reserved.              | -    | -     |
| 6:5   | GLITCH_DETECTOR_SENS   | RO   | 0x0   |
| 4     | GLITCH_DETECTOR_ENABLE | RO   | 0x0   |
| 3     | DEFAULT_ARCHSEL        | RO   | 0x0   |
| 2     | DEBUG_DISABLE          | RO   | 0x0   |
| 1     | SECURE_DEBUG_DISABLE   | RO   | 0x0   |
| 0     | SECURE_BOOT_ENABLE     | RO   | 0x0   |

## OTP: KEY\_VALID Register

Offset: 0x14c

Table 1351.  
KEY\_VALID Register

| Bits | Description | Type | Reset |
|------|-------------|------|-------|
| 31:8 | Reserved.   | -    | -     |

| Bits | Description                                   | Type | Reset |
|------|-----------------------------------------------|------|-------|
| 7:0  | Which keys were valid (enrolled) at boot time | RO   | 0x00  |

## OTP: DEBUGEN Register

Offset: 0x150

### Description

Enable a debug feature that has been disabled. Debug features are disabled if one of the relevant critical boot flags is set in OTP (DEBUG\_DISABLE or SECURE\_DEBUG\_DISABLE), OR if a debug key is marked valid in OTP, and the matching key value has not been supplied over SWD.

Specifically:

- The DEBUG\_DISABLE flag disables all debug features. This can be fully overridden by setting all bits of this register.
- The SECURE\_DEBUG\_DISABLE flag disables secure processor debug. This can be fully overridden by setting the PROC0\_SECURE and PROC1\_SECURE bits of this register.
- If a single debug key has been registered, and no matching key value has been supplied over SWD, then all debug features are disabled. This can be fully overridden by setting all bits of this register.
- If both debug keys have been registered, and the Non-secure key's value (key 6) has been supplied over SWD, secure processor debug is disabled. This can be fully overridden by setting the PROC0\_SECURE and PROC1\_SECURE bits of this register.
- If both debug keys have been registered, and the Secure key's value (key 5) has been supplied over SWD, then no debug features are disabled by the key mechanism. However, note that in this case debug features may still be disabled by the critical boot flags.

Table 1352. DEBUGEN Register

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                               | Type | Reset |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:9 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                 | -    | -     |
| 8    | <p><b>MISC:</b> Enable other debug components. Specifically, the CTI, and the APB-AP used to access the RISC-V Debug Module.</p> <p>These components are disabled by default if either of the debug disable critical flags is set, or if at least one debug key has been enrolled and the least secure of these enrolled key values has not been provided over SWD.</p>                                   | RW   | 0x0   |
| 7:4  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                 | -    | -     |
| 3    | <p><b>PROC1_SECURE:</b> Permit core 1's Mem-AP to generate Secure accesses, assuming it is enabled at all. Also enable secure debug of core 1 (SPIDEN and SPNIDEN).</p> <p>Secure debug of core 1 is disabled by default if the secure debug disable critical flag is set, or if at least one debug key has been enrolled and the most secure of these enrolled key values not yet provided over SWD.</p> | RW   | 0x0   |
| 2    | <p><b>PROC1:</b> Enable core 1's Mem-AP if it is currently disabled.</p> <p>The Mem-AP is disabled by default if either of the debug disable critical flags is set, or if at least one debug key has been enrolled and the least secure of these enrolled key values has not been provided over SWD.</p>                                                                                                  | RW   | 0x0   |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Type | Reset |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 1    | <p><b>PROC0_SECURE</b>: Permit core 0's Mem-AP to generate Secure accesses, assuming it is enabled at all. Also enable secure debug of core 0 (SPIDEN and SPNIDEN).</p> <p>Secure debug of core 0 is disabled by default if the secure debug disable critical flag is set, or if at least one debug key has been enrolled and the most secure of these enrolled key values not yet provided over SWD.</p> <p>Note also that core Mem-APs are unconditionally disabled when a core is switched to RISC-V mode (by setting the ARCHSEL bit and performing a warm reset of the core).</p> | RW   | 0x0   |
| 0    | <p><b>PROC0</b>: Enable core 0's Mem-AP if it is currently disabled.</p> <p>The Mem-AP is disabled by default if either of the debug disable critical flags is set, or if at least one debug key has been enrolled and the least secure of these enrolled key values has not been provided over SWD.</p> <p>Note also that core Mem-APs are unconditionally disabled when a core is switched to RISC-V mode (by setting the ARCHSEL bit and performing a warm reset of the core).</p>                                                                                                  | RW   | 0x0   |

## OTP: DEBUGEN\_LOCK Register

Offset: 0x154

### Description

Write 1s to lock corresponding bits in DEBUGEN. This register is reset by the processor cold reset.

Table 1353.  
DEBUGEN\_LOCK  
Register

| Bits | Description                                                                                       | Type | Reset |
|------|---------------------------------------------------------------------------------------------------|------|-------|
| 31:9 | Reserved.                                                                                         | -    | -     |
| 8    | <b>MISC</b> : Write 1 to lock the MISC bit of DEBUGEN. Can't be cleared once set.                 | RW   | 0x0   |
| 7:4  | Reserved.                                                                                         | -    | -     |
| 3    | <b>PROC1_SECURE</b> : Write 1 to lock the PROC1_SECURE bit of DEBUGEN. Can't be cleared once set. | RW   | 0x0   |
| 2    | <b>PROC1</b> : Write 1 to lock the PROC1 bit of DEBUGEN. Can't be cleared once set.               | RW   | 0x0   |
| 1    | <b>PROC0_SECURE</b> : Write 1 to lock the PROC0_SECURE bit of DEBUGEN. Can't be cleared once set. | RW   | 0x0   |
| 0    | <b>PROC0</b> : Write 1 to lock the PROC0 bit of DEBUGEN. Can't be cleared once set.               | RW   | 0x0   |

## OTP: ARCHSEL Register

Offset: 0x158

### Description

Architecture select (Arm/RISC-V). The default and allowable values of this register are constrained by the critical boot flags.

This register is reset by the earliest reset in the switched core power domain (before a processor cold reset).

Cores sample their architecture select signal on a warm reset. The source of the warm reset could be the system power-up state machine, the watchdog timer, Arm SYSRESETREQ or from RISC-V hartresetreq.

Note that when an Arm core is deselected, its cold reset domain is also held in reset, since in particular the

SYSRESETREQ bit becomes inaccessible once the core is deselected. Note also the RISC-V cores do not have a cold reset domain, since their corresponding controls are located in the Debug Module.

Table 1354. ARCHSEL Register

| Bits | Description                                     | Type | Reset |
|------|-------------------------------------------------|------|-------|
| 31:2 | Reserved.                                       | -    | -     |
| 1    | <b>CORE1</b> : Select architecture for core 1.  | RW   | 0x0   |
|      | Enumerated values:                              |      |       |
|      | 0x0 → ARM: Switch core 1 to Arm (Cortex-M33)    |      |       |
|      | 0x1 → RISC-V: Switch core 1 to RISC-V (Hazard3) |      |       |
| 0    | <b>CORE0</b> : Select architecture for core 0.  | RW   | 0x0   |
|      | Enumerated values:                              |      |       |
|      | 0x0 → ARM: Switch core 0 to Arm (Cortex-M33)    |      |       |
|      | 0x1 → RISC-V: Switch core 0 to RISC-V (Hazard3) |      |       |

## OTP: ARCHSEL\_STATUS Register

**Offset:** 0x15c

### Description

Get the current architecture select state of each core. Cores sample the current value of the ARCHSEL register when their warm reset is released, at which point the corresponding bit in this register will also update.

Table 1355. ARCHSEL\_STATUS Register

| Bits | Description                                                                      | Type | Reset |
|------|----------------------------------------------------------------------------------|------|-------|
| 31:2 | Reserved.                                                                        | -    | -     |
| 1    | <b>CORE1</b> : Current architecture for core 0. Updated on processor warm reset. | RO   | 0x0   |
|      | Enumerated values:                                                               |      |       |
|      | 0x0 → ARM: Core 1 is currently Arm (Cortex-M33)                                  |      |       |
|      | 0x1 → RISC-V: Core 1 is currently RISC-V (Hazard3)                               |      |       |
| 0    | <b>CORE0</b> : Current architecture for core 0. Updated on processor warm reset. | RO   | 0x0   |
|      | Enumerated values:                                                               |      |       |
|      | 0x0 → ARM: Core 0 is currently Arm (Cortex-M33)                                  |      |       |
|      | 0x1 → RISC-V: Core 0 is currently RISC-V (Hazard3)                               |      |       |

## OTP: BOOTDIS Register

**Offset:** 0x160

### Description

Tell the bootrom to ignore scratch register boot vectors (both power manager and watchdog) on the next power up.

If an early boot stage has soft-locked some OTP pages in order to protect their contents from later stages, there is a risk that Secure code running at a later stage can unlock the pages by performing a watchdog reset that resets the OTP.

This register can be used to ensure that the bootloader runs as normal on the next power up, preventing Secure code at a later stage from accessing OTP in its unlocked state.

Should be used in conjunction with the power manager BOOTDIS register.

Table 1356. BOOTDIS Register

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Type | Reset |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:2 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -    | -     |
| 1    | <p><b>NEXT:</b> This flag always ORs writes into its current contents. It can be set but not cleared by software.</p> <p>The BOOTDIS_NEXT bit is OR'd into the BOOTDIS_NOW bit when the core is powered down. Simultaneously, the BOOTDIS_NEXT bit is cleared. Setting this bit means that the boot scratch registers will be ignored following the next core power down.</p> <p>This flag should be set by an early boot stage that has soft-locked OTP pages, to prevent later stages from unlocking it via watchdog reset.</p> | RW   | 0x0   |
| 0    | <p><b>NOW:</b> When the core is powered down, the current value of BOOTDIS_NEXT is OR'd into BOOTDIS_NOW, and BOOTDIS_NEXT is cleared.</p> <p>The bootrom checks this flag before reading the boot scratch registers. If it is set, the bootrom clears it, and ignores the BOOT registers. This prevents Secure software from diverting the boot path before a bootloader has had the chance to soft lock OTP pages containing sensitive data.</p>                                                                                | WC   | 0x0   |

## OTP: INTR Register

Offset: 0x164

### Description

Raw Interrupts

Table 1357. INTR Register

| Bits | Description      | Type | Reset |
|------|------------------|------|-------|
| 31:5 | Reserved.        | -    | -     |
| 4    | APB_RD_NSEC_FAIL | WC   | 0x0   |
| 3    | APB_RD_SEC_FAIL  | WC   | 0x0   |
| 2    | APB_DCTRL_FAIL   | WC   | 0x0   |
| 1    | SBPI_WR_FAIL     | WC   | 0x0   |
| 0    | SBPI_FLAG_N      | RO   | 0x0   |

## OTP: INTE Register

Offset: 0x168

### Description

Interrupt Enable

Table 1358. INTE Register

| Bits | Description      | Type | Reset |
|------|------------------|------|-------|
| 31:5 | Reserved.        | -    | -     |
| 4    | APB_RD_NSEC_FAIL | RW   | 0x0   |
| 3    | APB_RD_SEC_FAIL  | RW   | 0x0   |
| 2    | APB_DCTRL_FAIL   | RW   | 0x0   |
| 1    | SBPI_WR_FAIL     | RW   | 0x0   |
| 0    | SBPI_FLAG_N      | RW   | 0x0   |

## OTP: INTF Register

Offset: 0x16c

### Description

Interrupt Force

Table 1359. INTF Register

| Bits | Description      | Type | Reset |
|------|------------------|------|-------|
| 31:5 | Reserved.        | -    | -     |
| 4    | APB_RD_NSEC_FAIL | RW   | 0x0   |
| 3    | APB_RD_SEC_FAIL  | RW   | 0x0   |
| 2    | APB_DCTRL_FAIL   | RW   | 0x0   |
| 1    | SBPI_WR_FAIL     | RW   | 0x0   |
| 0    | SBPI_FLAG_N      | RW   | 0x0   |

## OTP: INTS Register

Offset: 0x170

### Description

Interrupt status after masking & forcing

Table 1360. INTS Register

| Bits | Description      | Type | Reset |
|------|------------------|------|-------|
| 31:5 | Reserved.        | -    | -     |
| 4    | APB_RD_NSEC_FAIL | RO   | 0x0   |
| 3    | APB_RD_SEC_FAIL  | RO   | 0x0   |
| 2    | APB_DCTRL_FAIL   | RO   | 0x0   |
| 1    | SBPI_WR_FAIL     | RO   | 0x0   |
| 0    | SBPI_FLAG_N      | RO   | 0x0   |

## 13.9. Predefined OTP Data Locations

This section lists OTP locations used by either the hardware (particularly the OTP power-on state machine), the bootrom, or both. This listing is for RP2350 silicon revision A2.

OTP locations are listed by row number, not by address. When read through an ECC alias, OTP rows are spaced two bytes apart in the system address space; when read through a raw alias, OTP rows are four bytes apart. Therefore the row numbers given here should be multiplied by two or four appropriately when reading OTP contents directly from software. The OTP APIs provided by the bootrom use OTP row numbers directly, so this row-to-byte-address conversion is not necessary when accessing OTP through these APIs.

For normal (non-guarded) reads, you can access error-corrected content starting at [OTP\\_DATA\\_BASE](#) (0x40130000), and raw content starting at [OTP\\_DATA\\_RAW\\_BASE](#) (0x40134000). The register listings below indicate whether or not a given OTP row contains error-corrected contents. OTP never mixes error-corrected and non-error-corrected content in the same row.

All predefined data fields have some form of redundancy. Where ECC is not viable, for instance because a location is expected to have individual bits programmed at different times, best-of-three majority vote is used instead. The only exception to this is the critical hardware flags in [CRIT0](#) and [CRIT1](#). These flags use a three-of-eight vote encoding for each individual flag: the flag is considered set when at least three bits are set out of the eight redundant bit locations. The description for each row indicates the type of redundancy.

Pages 3 through 60 (rows `0x0c0` through `0xf3f`) are free for arbitrary user content such as OTP-resident bootloaders, and Raspberry Pi will avoid allocating any of these locations for bootrom configuration if possible. This is a total of 7424 bytes of ECC-protected content.

Page 2 (rows `0x080` through `0x0bf`) is also available for user content if secure boot is disabled. It is partially available if secure boot is enabled and fewer than four boot key fingerprints are registered. This is an additional 128 ECC bytes potentially available for user content.

Pages 0, 1, and 61 through 63 are reserved for future use by Raspberry Pi. Software should avoid allocating content in these regions, even if they currently have no defined use in this data listing.

Table 1361. List of  
OTP\_DATA registers

| Offset | Name       | Info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x000  | CHIPID0    | <p>Bits 15:0 of public device ID. (ECC)</p> <p>The CHIPID0..3 rows contain a 64-bit random identifier for this chip, which can be read from the USB bootloader PICOB00T interface or from the <code>get_sys_info</code> ROM API.</p> <p>The number of random bits makes the occurrence of twins exceedingly unlikely: for example, a fleet of a hundred million devices has a 99.97% probability of no twinned IDs. This is estimated to be lower than the occurrence of process errors in the assignment of sequential random IDs, and for practical purposes CHIPID may be treated as unique.</p> |
| 0x001  | CHIPID1    | Bits 31:16 of public device ID (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x002  | CHIPID2    | Bits 47:32 of public device ID (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x003  | CHIPID3    | Bits 63:48 of public device ID (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x004  | RANDID0    | <p>Bits 15:0 of private per-device random number (ECC)</p> <p>The RANDID0..7 rows form a 128-bit random number generated during device test.</p> <p>This ID is not exposed through the USB PICOB00T GET_INFO command or the ROM <code>get_sys_info()</code> API. However note that the USB PICOB00T OTP access point can read the entirety of page 0, so this value is not meaningfully private unless the USB PICOB00T interface is disabled via the <code>DISABLE_BOOTSEL_USB_PICOB00T_IFC</code> flag in <code>BOOT_FLAGS0</code>.</p>                                                           |
| 0x005  | RANDID1    | Bits 31:16 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x006  | RANDID2    | Bits 47:32 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x007  | RANDID3    | Bits 63:48 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x008  | RANDID4    | Bits 79:64 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x009  | RANDID5    | Bits 95:80 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0x00a  | RANDID6    | Bits 111:96 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0x00b  | RANDID7    | Bits 127:112 of private per-device random number (ECC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0x010  | ROSC_CALIB | <p>Ring oscillator frequency in kHz, measured during manufacturing (ECC)</p> <p>This is measured at 1.1 V, at room temperature, with the ROSC configuration registers in their reset state.</p>                                                                                                                                                                                                                                                                                                                                                                                                     |

| Offset | Name                     | Info                                                                                                                                                                                        |
|--------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x011  | LPOSC_CALIB              | Low-power oscillator frequency in Hz, measured during manufacturing (ECC)<br><br>This is measured at 1.1V, at room temperature, with the LPOSC trim register in its reset state.            |
| 0x018  | NUM_GPIOS                | The number of main user GPIOs (bank 0). Should read 48 in the QFN80 package, and 30 in the QFN60 package. (ECC)                                                                             |
| 0x036  | INFO_CRC0                | Lower 16 bits of CRC32 of OTP addresses 0x00 through 0x6b (polynomial 0x4c11db7, input reflected, output reflected, seed all-ones, final XOR all-ones) (ECC)                                |
| 0x037  | INFO_CRC1                | Upper 16 bits of CRC32 of OTP addresses 0x00 through 0x6b (ECC)                                                                                                                             |
| 0x038  | CRIT0                    | Page 0 critical boot flags (RBIT-8)                                                                                                                                                         |
| 0x039  | CRIT0_R1                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x03a  | CRIT0_R2                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x03b  | CRIT0_R3                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x03c  | CRIT0_R4                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x03d  | CRIT0_R5                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x03e  | CRIT0_R6                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x03f  | CRIT0_R7                 | Redundant copy of CRIT0                                                                                                                                                                     |
| 0x040  | CRIT1                    | Page 1 critical boot flags (RBIT-8)                                                                                                                                                         |
| 0x041  | CRIT1_R1                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x042  | CRIT1_R2                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x043  | CRIT1_R3                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x044  | CRIT1_R4                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x045  | CRIT1_R5                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x046  | CRIT1_R6                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x047  | CRIT1_R7                 | Redundant copy of CRIT1                                                                                                                                                                     |
| 0x048  | BOOT_FLAGS0              | Disable/Enable boot paths/features in the RP2350 mask ROM. Disables always supersede enables. Enables are provided where there are other configurations in OTP that must be valid. (RBIT-3) |
| 0x049  | BOOT_FLAGS0_R1           | Redundant copy of BOOT_FLAGS0                                                                                                                                                               |
| 0x04a  | BOOT_FLAGS0_R2           | Redundant copy of BOOT_FLAGS0                                                                                                                                                               |
| 0x04b  | BOOT_FLAGS1              | Disable/Enable boot paths/features in the RP2350 mask ROM. Disables always supersede enables. Enables are provided where there are other configurations in OTP that must be valid. (RBIT-3) |
| 0x04c  | BOOT_FLAGS1_R1           | Redundant copy of BOOT_FLAGS1                                                                                                                                                               |
| 0x04d  | BOOT_FLAGS1_R2           | Redundant copy of BOOT_FLAGS1                                                                                                                                                               |
| 0x04e  | DEFAULT_BOOT_VERSION0    | Default boot version thermometer counter, bits 23:0 (RBIT-3)                                                                                                                                |
| 0x04f  | DEFAULT_BOOT_VERSION0_R1 | Redundant copy of DEFAULT_BOOT_VERSION0                                                                                                                                                     |

| Offset | Name                                      | Info                                                                                                                                                                                                                 |
|--------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x050  | <a href="#">DEFAULT_BOOT_VERSION0_R2</a>  | Redundant copy of DEFAULT_BOOT_VERSION0                                                                                                                                                                              |
| 0x051  | <a href="#">DEFAULT_BOOT_VERSION1</a>     | Default boot version thermometer counter, bits 47:24 (RBIT-3)                                                                                                                                                        |
| 0x052  | <a href="#">DEFAULT_BOOT_VERSION1_R1</a>  | Redundant copy of DEFAULT_BOOT_VERSION1                                                                                                                                                                              |
| 0x053  | <a href="#">DEFAULT_BOOT_VERSION1_R2</a>  | Redundant copy of DEFAULT_BOOT_VERSION1                                                                                                                                                                              |
| 0x054  | <a href="#">FLASH_DEVINFO</a>             | Stores information about external flash device(s). (ECC)<br><br>Assumed to be valid if<br>BOOT_FLAGS0_FLASH_DEVINFO_ENABLE is set.                                                                                   |
| 0x055  | <a href="#">FLASH_PARTITION_SLOT_SIZE</a> | Gap between partition table slot 0 and slot 1 at the start of flash (the default size is 4096 bytes) (ECC) Enabled by the<br>OVERRIDE_FLASH_PARTITION_SLOT_SIZE bit in BOOT_FLAGS,<br>the size is 4096 * (value + 1) |
| 0x056  | <a href="#">BOOTSEL_LED_CFG</a>           | Pin configuration for LED status, used by USB bootloader. (ECC)<br>Must be valid if BOOT_FLAGS0_ENABLE_BOOTSEL_LED is set.                                                                                           |
| 0x057  | <a href="#">BOOTSEL_PLL_CFG</a>           | Optional PLL configuration for BOOTSEL mode. (ECC)                                                                                                                                                                   |
| 0x058  | <a href="#">BOOTSEL_XOSC_CFG</a>          | Non-default crystal oscillator configuration for the USB<br>bootloader. (ECC)                                                                                                                                        |
| 0x059  | <a href="#">USB_BOOT_FLAGS</a>            | USB boot specific feature flags (RBIT-3)                                                                                                                                                                             |
| 0x05a  | <a href="#">USB_BOOT_FLAGS_R1</a>         | Redundant copy of USB_BOOT_FLAGS                                                                                                                                                                                     |
| 0x05b  | <a href="#">USB_BOOT_FLAGS_R2</a>         | Redundant copy of USB_BOOT_FLAGS                                                                                                                                                                                     |
| 0x05c  | <a href="#">USB_WHITE_LABEL_ADDR</a>      | Row index of the USB_WHITE_LABEL structure within OTP (ECC)                                                                                                                                                          |
| 0x05e  | <a href="#">OTPBOOT_SRC</a>               | OTP start row for the OTP boot image. (ECC)                                                                                                                                                                          |
| 0x05f  | <a href="#">OTPBOOT_LEN</a>               | Length in rows of the OTP boot image. (ECC)                                                                                                                                                                          |
| 0x060  | <a href="#">OTPBOOT_DST0</a>              | Bits 15:0 of the OTP boot image load destination (and entry<br>point). (ECC)                                                                                                                                         |
| 0x061  | <a href="#">OTPBOOT_DST1</a>              | Bits 31:16 of the OTP boot image load destination (and entry<br>point). (ECC)                                                                                                                                        |
| 0x080  | <a href="#">BOOTKEY0_0</a>                | Bits 15:0 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                        |
| 0x081  | <a href="#">BOOTKEY0_1</a>                | Bits 31:16 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                       |
| 0x082  | <a href="#">BOOTKEY0_2</a>                | Bits 47:32 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                       |
| 0x083  | <a href="#">BOOTKEY0_3</a>                | Bits 63:48 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                       |
| 0x084  | <a href="#">BOOTKEY0_4</a>                | Bits 79:64 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                       |
| 0x085  | <a href="#">BOOTKEY0_5</a>                | Bits 95:80 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                       |
| 0x086  | <a href="#">BOOTKEY0_6</a>                | Bits 111:96 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                      |
| 0x087  | <a href="#">BOOTKEY0_7</a>                | Bits 127:112 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                     |
| 0x088  | <a href="#">BOOTKEY0_8</a>                | Bits 143:128 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                     |
| 0x089  | <a href="#">BOOTKEY0_9</a>                | Bits 159:144 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                     |
| 0x08a  | <a href="#">BOOTKEY0_10</a>               | Bits 175:160 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                     |
| 0x08b  | <a href="#">BOOTKEY0_11</a>               | Bits 191:176 of SHA-256 hash of boot key 0 (ECC)                                                                                                                                                                     |

| Offset | Name                        | Info                                             |
|--------|-----------------------------|--------------------------------------------------|
| 0x08c  | <a href="#">BOOTKEY0_12</a> | Bits 207:192 of SHA-256 hash of boot key 0 (ECC) |
| 0x08d  | <a href="#">BOOTKEY0_13</a> | Bits 223:208 of SHA-256 hash of boot key 0 (ECC) |
| 0x08e  | <a href="#">BOOTKEY0_14</a> | Bits 239:224 of SHA-256 hash of boot key 0 (ECC) |
| 0x08f  | <a href="#">BOOTKEY0_15</a> | Bits 255:240 of SHA-256 hash of boot key 0 (ECC) |
| 0x090  | <a href="#">BOOTKEY1_0</a>  | Bits 15:0 of SHA-256 hash of boot key 1 (ECC)    |
| 0x091  | <a href="#">BOOTKEY1_1</a>  | Bits 31:16 of SHA-256 hash of boot key 1 (ECC)   |
| 0x092  | <a href="#">BOOTKEY1_2</a>  | Bits 47:32 of SHA-256 hash of boot key 1 (ECC)   |
| 0x093  | <a href="#">BOOTKEY1_3</a>  | Bits 63:48 of SHA-256 hash of boot key 1 (ECC)   |
| 0x094  | <a href="#">BOOTKEY1_4</a>  | Bits 79:64 of SHA-256 hash of boot key 1 (ECC)   |
| 0x095  | <a href="#">BOOTKEY1_5</a>  | Bits 95:80 of SHA-256 hash of boot key 1 (ECC)   |
| 0x096  | <a href="#">BOOTKEY1_6</a>  | Bits 111:96 of SHA-256 hash of boot key 1 (ECC)  |
| 0x097  | <a href="#">BOOTKEY1_7</a>  | Bits 127:112 of SHA-256 hash of boot key 1 (ECC) |
| 0x098  | <a href="#">BOOTKEY1_8</a>  | Bits 143:128 of SHA-256 hash of boot key 1 (ECC) |
| 0x099  | <a href="#">BOOTKEY1_9</a>  | Bits 159:144 of SHA-256 hash of boot key 1 (ECC) |
| 0x09a  | <a href="#">BOOTKEY1_10</a> | Bits 175:160 of SHA-256 hash of boot key 1 (ECC) |
| 0x09b  | <a href="#">BOOTKEY1_11</a> | Bits 191:176 of SHA-256 hash of boot key 1 (ECC) |
| 0x09c  | <a href="#">BOOTKEY1_12</a> | Bits 207:192 of SHA-256 hash of boot key 1 (ECC) |
| 0x09d  | <a href="#">BOOTKEY1_13</a> | Bits 223:208 of SHA-256 hash of boot key 1 (ECC) |
| 0x09e  | <a href="#">BOOTKEY1_14</a> | Bits 239:224 of SHA-256 hash of boot key 1 (ECC) |
| 0x09f  | <a href="#">BOOTKEY1_15</a> | Bits 255:240 of SHA-256 hash of boot key 1 (ECC) |
| 0x0a0  | <a href="#">BOOTKEY2_0</a>  | Bits 15:0 of SHA-256 hash of boot key 2 (ECC)    |
| 0x0a1  | <a href="#">BOOTKEY2_1</a>  | Bits 31:16 of SHA-256 hash of boot key 2 (ECC)   |
| 0x0a2  | <a href="#">BOOTKEY2_2</a>  | Bits 47:32 of SHA-256 hash of boot key 2 (ECC)   |
| 0x0a3  | <a href="#">BOOTKEY2_3</a>  | Bits 63:48 of SHA-256 hash of boot key 2 (ECC)   |
| 0x0a4  | <a href="#">BOOTKEY2_4</a>  | Bits 79:64 of SHA-256 hash of boot key 2 (ECC)   |
| 0x0a5  | <a href="#">BOOTKEY2_5</a>  | Bits 95:80 of SHA-256 hash of boot key 2 (ECC)   |
| 0x0a6  | <a href="#">BOOTKEY2_6</a>  | Bits 111:96 of SHA-256 hash of boot key 2 (ECC)  |
| 0x0a7  | <a href="#">BOOTKEY2_7</a>  | Bits 127:112 of SHA-256 hash of boot key 2 (ECC) |
| 0x0a8  | <a href="#">BOOTKEY2_8</a>  | Bits 143:128 of SHA-256 hash of boot key 2 (ECC) |
| 0x0a9  | <a href="#">BOOTKEY2_9</a>  | Bits 159:144 of SHA-256 hash of boot key 2 (ECC) |
| 0x0aa  | <a href="#">BOOTKEY2_10</a> | Bits 175:160 of SHA-256 hash of boot key 2 (ECC) |
| 0x0ab  | <a href="#">BOOTKEY2_11</a> | Bits 191:176 of SHA-256 hash of boot key 2 (ECC) |
| 0x0ac  | <a href="#">BOOTKEY2_12</a> | Bits 207:192 of SHA-256 hash of boot key 2 (ECC) |
| 0x0ad  | <a href="#">BOOTKEY2_13</a> | Bits 223:208 of SHA-256 hash of boot key 2 (ECC) |
| 0x0ae  | <a href="#">BOOTKEY2_14</a> | Bits 239:224 of SHA-256 hash of boot key 2 (ECC) |
| 0x0af  | <a href="#">BOOTKEY2_15</a> | Bits 255:240 of SHA-256 hash of boot key 2 (ECC) |

| Offset | Name                        | Info                                             |
|--------|-----------------------------|--------------------------------------------------|
| 0x0b0  | <a href="#">BOOTKEY3_0</a>  | Bits 15:0 of SHA-256 hash of boot key 3 (ECC)    |
| 0x0b1  | <a href="#">BOOTKEY3_1</a>  | Bits 31:16 of SHA-256 hash of boot key 3 (ECC)   |
| 0x0b2  | <a href="#">BOOTKEY3_2</a>  | Bits 47:32 of SHA-256 hash of boot key 3 (ECC)   |
| 0x0b3  | <a href="#">BOOTKEY3_3</a>  | Bits 63:48 of SHA-256 hash of boot key 3 (ECC)   |
| 0x0b4  | <a href="#">BOOTKEY3_4</a>  | Bits 79:64 of SHA-256 hash of boot key 3 (ECC)   |
| 0x0b5  | <a href="#">BOOTKEY3_5</a>  | Bits 95:80 of SHA-256 hash of boot key 3 (ECC)   |
| 0x0b6  | <a href="#">BOOTKEY3_6</a>  | Bits 111:96 of SHA-256 hash of boot key 3 (ECC)  |
| 0x0b7  | <a href="#">BOOTKEY3_7</a>  | Bits 127:112 of SHA-256 hash of boot key 3 (ECC) |
| 0x0b8  | <a href="#">BOOTKEY3_8</a>  | Bits 143:128 of SHA-256 hash of boot key 3 (ECC) |
| 0x0b9  | <a href="#">BOOTKEY3_9</a>  | Bits 159:144 of SHA-256 hash of boot key 3 (ECC) |
| 0x0ba  | <a href="#">BOOTKEY3_10</a> | Bits 175:160 of SHA-256 hash of boot key 3 (ECC) |
| 0x0bb  | <a href="#">BOOTKEY3_11</a> | Bits 191:176 of SHA-256 hash of boot key 3 (ECC) |
| 0x0bc  | <a href="#">BOOTKEY3_12</a> | Bits 207:192 of SHA-256 hash of boot key 3 (ECC) |
| 0x0bd  | <a href="#">BOOTKEY3_13</a> | Bits 223:208 of SHA-256 hash of boot key 3 (ECC) |
| 0x0be  | <a href="#">BOOTKEY3_14</a> | Bits 239:224 of SHA-256 hash of boot key 3 (ECC) |
| 0x0bf  | <a href="#">BOOTKEY3_15</a> | Bits 255:240 of SHA-256 hash of boot key 3 (ECC) |
| 0xf48  | <a href="#">KEY1_0</a>      | Bits 15:0 of OTP access key 1 (ECC)              |
| 0xf49  | <a href="#">KEY1_1</a>      | Bits 31:16 of OTP access key 1 (ECC)             |
| 0xf4a  | <a href="#">KEY1_2</a>      | Bits 47:32 of OTP access key 1 (ECC)             |
| 0xf4b  | <a href="#">KEY1_3</a>      | Bits 63:48 of OTP access key 1 (ECC)             |
| 0xf4c  | <a href="#">KEY1_4</a>      | Bits 79:64 of OTP access key 1 (ECC)             |
| 0xf4d  | <a href="#">KEY1_5</a>      | Bits 95:80 of OTP access key 1 (ECC)             |
| 0xf4e  | <a href="#">KEY1_6</a>      | Bits 111:96 of OTP access key 1 (ECC)            |
| 0xf4f  | <a href="#">KEY1_7</a>      | Bits 127:112 of OTP access key 1 (ECC)           |
| 0xf50  | <a href="#">KEY2_0</a>      | Bits 15:0 of OTP access key 2 (ECC)              |
| 0xf51  | <a href="#">KEY2_1</a>      | Bits 31:16 of OTP access key 2 (ECC)             |
| 0xf52  | <a href="#">KEY2_2</a>      | Bits 47:32 of OTP access key 2 (ECC)             |
| 0xf53  | <a href="#">KEY2_3</a>      | Bits 63:48 of OTP access key 2 (ECC)             |
| 0xf54  | <a href="#">KEY2_4</a>      | Bits 79:64 of OTP access key 2 (ECC)             |
| 0xf55  | <a href="#">KEY2_5</a>      | Bits 95:80 of OTP access key 2 (ECC)             |
| 0xf56  | <a href="#">KEY2_6</a>      | Bits 111:96 of OTP access key 2 (ECC)            |
| 0xf57  | <a href="#">KEY2_7</a>      | Bits 127:112 of OTP access key 2 (ECC)           |
| 0xf58  | <a href="#">KEY3_0</a>      | Bits 15:0 of OTP access key 3 (ECC)              |
| 0xf59  | <a href="#">KEY3_1</a>      | Bits 31:16 of OTP access key 3 (ECC)             |
| 0xf5a  | <a href="#">KEY3_2</a>      | Bits 47:32 of OTP access key 3 (ECC)             |
| 0xf5b  | <a href="#">KEY3_3</a>      | Bits 63:48 of OTP access key 3 (ECC)             |

| Offset | Name                        | Info                                                        |
|--------|-----------------------------|-------------------------------------------------------------|
| 0xf5c  | <a href="#">KEY3_4</a>      | Bits 79:64 of OTP access key 3 (ECC)                        |
| 0xf5d  | <a href="#">KEY3_5</a>      | Bits 95:80 of OTP access key 3 (ECC)                        |
| 0xf5e  | <a href="#">KEY3_6</a>      | Bits 111:96 of OTP access key 3 (ECC)                       |
| 0xf5f  | <a href="#">KEY3_7</a>      | Bits 127:112 of OTP access key 3 (ECC)                      |
| 0xf60  | <a href="#">KEY4_0</a>      | Bits 15:0 of OTP access key 4 (ECC)                         |
| 0xf61  | <a href="#">KEY4_1</a>      | Bits 31:16 of OTP access key 4 (ECC)                        |
| 0xf62  | <a href="#">KEY4_2</a>      | Bits 47:32 of OTP access key 4 (ECC)                        |
| 0xf63  | <a href="#">KEY4_3</a>      | Bits 63:48 of OTP access key 4 (ECC)                        |
| 0xf64  | <a href="#">KEY4_4</a>      | Bits 79:64 of OTP access key 4 (ECC)                        |
| 0xf65  | <a href="#">KEY4_5</a>      | Bits 95:80 of OTP access key 4 (ECC)                        |
| 0xf66  | <a href="#">KEY4_6</a>      | Bits 111:96 of OTP access key 4 (ECC)                       |
| 0xf67  | <a href="#">KEY4_7</a>      | Bits 127:112 of OTP access key 4 (ECC)                      |
| 0xf68  | <a href="#">KEY5_0</a>      | Bits 15:0 of OTP access key 5 (ECC)                         |
| 0xf69  | <a href="#">KEY5_1</a>      | Bits 31:16 of OTP access key 5 (ECC)                        |
| 0xf6a  | <a href="#">KEY5_2</a>      | Bits 47:32 of OTP access key 5 (ECC)                        |
| 0xf6b  | <a href="#">KEY5_3</a>      | Bits 63:48 of OTP access key 5 (ECC)                        |
| 0xf6c  | <a href="#">KEY5_4</a>      | Bits 79:64 of OTP access key 5 (ECC)                        |
| 0xf6d  | <a href="#">KEY5_5</a>      | Bits 95:80 of OTP access key 5 (ECC)                        |
| 0xf6e  | <a href="#">KEY5_6</a>      | Bits 111:96 of OTP access key 5 (ECC)                       |
| 0xf6f  | <a href="#">KEY5_7</a>      | Bits 127:112 of OTP access key 5 (ECC)                      |
| 0xf70  | <a href="#">KEY6_0</a>      | Bits 15:0 of OTP access key 6 (ECC)                         |
| 0xf71  | <a href="#">KEY6_1</a>      | Bits 31:16 of OTP access key 6 (ECC)                        |
| 0xf72  | <a href="#">KEY6_2</a>      | Bits 47:32 of OTP access key 6 (ECC)                        |
| 0xf73  | <a href="#">KEY6_3</a>      | Bits 63:48 of OTP access key 6 (ECC)                        |
| 0xf74  | <a href="#">KEY6_4</a>      | Bits 79:64 of OTP access key 6 (ECC)                        |
| 0xf75  | <a href="#">KEY6_5</a>      | Bits 95:80 of OTP access key 6 (ECC)                        |
| 0xf76  | <a href="#">KEY6_6</a>      | Bits 111:96 of OTP access key 6 (ECC)                       |
| 0xf77  | <a href="#">KEY6_7</a>      | Bits 127:112 of OTP access key 6 (ECC)                      |
| 0xf79  | <a href="#">KEY1_VALID</a>  | Valid flag for key 1.                                       |
| 0xf7a  | <a href="#">KEY2_VALID</a>  | Valid flag for key 2.                                       |
| 0xf7b  | <a href="#">KEY3_VALID</a>  | Valid flag for key 3.                                       |
| 0xf7c  | <a href="#">KEY4_VALID</a>  | Valid flag for key 4.                                       |
| 0xf7d  | <a href="#">KEY5_VALID</a>  | Valid flag for key 5.                                       |
| 0xf7e  | <a href="#">KEY6_VALID</a>  | Valid flag for key 6.                                       |
| 0xf80  | <a href="#">PAGE0_LOCK0</a> | Lock configuration LSBs for page 0 (rows 0x0 through 0x3f). |
| 0xf81  | <a href="#">PAGE0_LOCK1</a> | Lock configuration MSBs for page 0 (rows 0x0 through 0x3f). |

| Offset | Name                         | Info                                                            |
|--------|------------------------------|-----------------------------------------------------------------|
| 0xf82  | <a href="#">PAGE1_LOCK0</a>  | Lock configuration LSBs for page 1 (rows 0x40 through 0x7f).    |
| 0xf83  | <a href="#">PAGE1_LOCK1</a>  | Lock configuration MSBs for page 1 (rows 0x40 through 0x7f).    |
| 0xf84  | <a href="#">PAGE2_LOCK0</a>  | Lock configuration LSBs for page 2 (rows 0x80 through 0xbf).    |
| 0xf85  | <a href="#">PAGE2_LOCK1</a>  | Lock configuration MSBs for page 2 (rows 0x80 through 0xbf).    |
| 0xf86  | <a href="#">PAGE3_LOCK0</a>  | Lock configuration LSBs for page 3 (rows 0xc0 through 0xff).    |
| 0xf87  | <a href="#">PAGE3_LOCK1</a>  | Lock configuration MSBs for page 3 (rows 0xc0 through 0xff).    |
| 0xf88  | <a href="#">PAGE4_LOCK0</a>  | Lock configuration LSBs for page 4 (rows 0x100 through 0x13f).  |
| 0xf89  | <a href="#">PAGE4_LOCK1</a>  | Lock configuration MSBs for page 4 (rows 0x100 through 0x13f).  |
| 0xf8a  | <a href="#">PAGE5_LOCK0</a>  | Lock configuration LSBs for page 5 (rows 0x140 through 0x17f).  |
| 0xf8b  | <a href="#">PAGE5_LOCK1</a>  | Lock configuration MSBs for page 5 (rows 0x140 through 0x17f).  |
| 0xf8c  | <a href="#">PAGE6_LOCK0</a>  | Lock configuration LSBs for page 6 (rows 0x180 through 0x1bf).  |
| 0xf8d  | <a href="#">PAGE6_LOCK1</a>  | Lock configuration MSBs for page 6 (rows 0x180 through 0x1bf).  |
| 0xf8e  | <a href="#">PAGE7_LOCK0</a>  | Lock configuration LSBs for page 7 (rows 0x1c0 through 0x1ff).  |
| 0xf8f  | <a href="#">PAGE7_LOCK1</a>  | Lock configuration MSBs for page 7 (rows 0x1c0 through 0x1ff).  |
| 0xf90  | <a href="#">PAGE8_LOCK0</a>  | Lock configuration LSBs for page 8 (rows 0x200 through 0x23f).  |
| 0xf91  | <a href="#">PAGE8_LOCK1</a>  | Lock configuration MSBs for page 8 (rows 0x200 through 0x23f).  |
| 0xf92  | <a href="#">PAGE9_LOCK0</a>  | Lock configuration LSBs for page 9 (rows 0x240 through 0x27f).  |
| 0xf93  | <a href="#">PAGE9_LOCK1</a>  | Lock configuration MSBs for page 9 (rows 0x240 through 0x27f).  |
| 0xf94  | <a href="#">PAGE10_LOCK0</a> | Lock configuration LSBs for page 10 (rows 0x280 through 0x2bf). |
| 0xf95  | <a href="#">PAGE10_LOCK1</a> | Lock configuration MSBs for page 10 (rows 0x280 through 0x2bf). |
| 0xf96  | <a href="#">PAGE11_LOCK0</a> | Lock configuration LSBs for page 11 (rows 0x2c0 through 0x2ff). |
| 0xf97  | <a href="#">PAGE11_LOCK1</a> | Lock configuration MSBs for page 11 (rows 0x2c0 through 0x2ff). |
| 0xf98  | <a href="#">PAGE12_LOCK0</a> | Lock configuration LSBs for page 12 (rows 0x300 through 0x33f). |
| 0xf99  | <a href="#">PAGE12_LOCK1</a> | Lock configuration MSBs for page 12 (rows 0x300 through 0x33f). |
| 0xf9a  | <a href="#">PAGE13_LOCK0</a> | Lock configuration LSBs for page 13 (rows 0x340 through 0x37f). |
| 0xf9b  | <a href="#">PAGE13_LOCK1</a> | Lock configuration MSBs for page 13 (rows 0x340 through 0x37f). |
| 0xf9c  | <a href="#">PAGE14_LOCK0</a> | Lock configuration LSBs for page 14 (rows 0x380 through 0x3bf). |
| 0xf9d  | <a href="#">PAGE14_LOCK1</a> | Lock configuration MSBs for page 14 (rows 0x380 through 0x3bf). |
| 0xf9e  | <a href="#">PAGE15_LOCK0</a> | Lock configuration LSBs for page 15 (rows 0x3c0 through 0x3ff). |

| Offset | Name                         | Info                                                            |
|--------|------------------------------|-----------------------------------------------------------------|
| 0xf9f  | <a href="#">PAGE15_LOCK1</a> | Lock configuration MSBs for page 15 (rows 0x3c0 through 0x3ff). |
| 0xfa0  | <a href="#">PAGE16_LOCK0</a> | Lock configuration LSBs for page 16 (rows 0x400 through 0x43f). |
| 0xfa1  | <a href="#">PAGE16_LOCK1</a> | Lock configuration MSBs for page 16 (rows 0x400 through 0x43f). |
| 0xfa2  | <a href="#">PAGE17_LOCK0</a> | Lock configuration LSBs for page 17 (rows 0x440 through 0x47f). |
| 0xfa3  | <a href="#">PAGE17_LOCK1</a> | Lock configuration MSBs for page 17 (rows 0x440 through 0x47f). |
| 0xfa4  | <a href="#">PAGE18_LOCK0</a> | Lock configuration LSBs for page 18 (rows 0x480 through 0x4bf). |
| 0xfa5  | <a href="#">PAGE18_LOCK1</a> | Lock configuration MSBs for page 18 (rows 0x480 through 0x4bf). |
| 0xfa6  | <a href="#">PAGE19_LOCK0</a> | Lock configuration LSBs for page 19 (rows 0x4c0 through 0x4ff). |
| 0xfa7  | <a href="#">PAGE19_LOCK1</a> | Lock configuration MSBs for page 19 (rows 0x4c0 through 0x4ff). |
| 0xfa8  | <a href="#">PAGE20_LOCK0</a> | Lock configuration LSBs for page 20 (rows 0x500 through 0x53f). |
| 0xfa9  | <a href="#">PAGE20_LOCK1</a> | Lock configuration MSBs for page 20 (rows 0x500 through 0x53f). |
| 0xfaa  | <a href="#">PAGE21_LOCK0</a> | Lock configuration LSBs for page 21 (rows 0x540 through 0x57f). |
| 0xfab  | <a href="#">PAGE21_LOCK1</a> | Lock configuration MSBs for page 21 (rows 0x540 through 0x57f). |
| 0xfac  | <a href="#">PAGE22_LOCK0</a> | Lock configuration LSBs for page 22 (rows 0x580 through 0x5bf). |
| 0xfad  | <a href="#">PAGE22_LOCK1</a> | Lock configuration MSBs for page 22 (rows 0x580 through 0x5bf). |
| 0xfae  | <a href="#">PAGE23_LOCK0</a> | Lock configuration LSBs for page 23 (rows 0x5c0 through 0x5ff). |
| 0xfaf  | <a href="#">PAGE23_LOCK1</a> | Lock configuration MSBs for page 23 (rows 0x5c0 through 0x5ff). |
| 0xfb0  | <a href="#">PAGE24_LOCK0</a> | Lock configuration LSBs for page 24 (rows 0x600 through 0x63f). |
| 0xfb1  | <a href="#">PAGE24_LOCK1</a> | Lock configuration MSBs for page 24 (rows 0x600 through 0x63f). |
| 0xfb2  | <a href="#">PAGE25_LOCK0</a> | Lock configuration LSBs for page 25 (rows 0x640 through 0x67f). |
| 0xfb3  | <a href="#">PAGE25_LOCK1</a> | Lock configuration MSBs for page 25 (rows 0x640 through 0x67f). |
| 0xfb4  | <a href="#">PAGE26_LOCK0</a> | Lock configuration LSBs for page 26 (rows 0x680 through 0x6bf). |

| Offset | Name                         | Info                                                            |
|--------|------------------------------|-----------------------------------------------------------------|
| 0xfb5  | <a href="#">PAGE26_LOCK1</a> | Lock configuration MSBs for page 26 (rows 0x680 through 0x6bf). |
| 0xfb6  | <a href="#">PAGE27_LOCK0</a> | Lock configuration LSBs for page 27 (rows 0x6c0 through 0x6ff). |
| 0xfb7  | <a href="#">PAGE27_LOCK1</a> | Lock configuration MSBs for page 27 (rows 0x6c0 through 0x6ff). |
| 0xfb8  | <a href="#">PAGE28_LOCK0</a> | Lock configuration LSBs for page 28 (rows 0x700 through 0x73f). |
| 0xfb9  | <a href="#">PAGE28_LOCK1</a> | Lock configuration MSBs for page 28 (rows 0x700 through 0x73f). |
| 0xfba  | <a href="#">PAGE29_LOCK0</a> | Lock configuration LSBs for page 29 (rows 0x740 through 0x77f). |
| 0xfbb  | <a href="#">PAGE29_LOCK1</a> | Lock configuration MSBs for page 29 (rows 0x740 through 0x77f). |
| 0xfbc  | <a href="#">PAGE30_LOCK0</a> | Lock configuration LSBs for page 30 (rows 0x780 through 0x7bf). |
| 0xfbd  | <a href="#">PAGE30_LOCK1</a> | Lock configuration MSBs for page 30 (rows 0x780 through 0x7bf). |
| 0xfbe  | <a href="#">PAGE31_LOCK0</a> | Lock configuration LSBs for page 31 (rows 0x7c0 through 0x7ff). |
| 0xfbfb | <a href="#">PAGE31_LOCK1</a> | Lock configuration MSBs for page 31 (rows 0x7c0 through 0x7ff). |
| 0xfc0  | <a href="#">PAGE32_LOCK0</a> | Lock configuration LSBs for page 32 (rows 0x800 through 0x83f). |
| 0xfc1  | <a href="#">PAGE32_LOCK1</a> | Lock configuration MSBs for page 32 (rows 0x800 through 0x83f). |
| 0xfc2  | <a href="#">PAGE33_LOCK0</a> | Lock configuration LSBs for page 33 (rows 0x840 through 0x87f). |
| 0xfc3  | <a href="#">PAGE33_LOCK1</a> | Lock configuration MSBs for page 33 (rows 0x840 through 0x87f). |
| 0xfc4  | <a href="#">PAGE34_LOCK0</a> | Lock configuration LSBs for page 34 (rows 0x880 through 0x8bf). |
| 0xfc5  | <a href="#">PAGE34_LOCK1</a> | Lock configuration MSBs for page 34 (rows 0x880 through 0x8bf). |
| 0xfc6  | <a href="#">PAGE35_LOCK0</a> | Lock configuration LSBs for page 35 (rows 0x8c0 through 0x8ff). |
| 0xfc7  | <a href="#">PAGE35_LOCK1</a> | Lock configuration MSBs for page 35 (rows 0x8c0 through 0x8ff). |
| 0xfc8  | <a href="#">PAGE36_LOCK0</a> | Lock configuration LSBs for page 36 (rows 0x900 through 0x93f). |
| 0xfc9  | <a href="#">PAGE36_LOCK1</a> | Lock configuration MSBs for page 36 (rows 0x900 through 0x93f). |
| 0xfca  | <a href="#">PAGE37_LOCK0</a> | Lock configuration LSBs for page 37 (rows 0x940 through 0x97f). |
| 0xfcb  | <a href="#">PAGE37_LOCK1</a> | Lock configuration MSBs for page 37 (rows 0x940 through 0x97f). |

| Offset | Name                         | Info                                                            |
|--------|------------------------------|-----------------------------------------------------------------|
| 0xfcc  | <a href="#">PAGE38_LOCK0</a> | Lock configuration LSBs for page 38 (rows 0x980 through 0x9bf). |
| 0xfcd  | <a href="#">PAGE38_LOCK1</a> | Lock configuration MSBs for page 38 (rows 0x980 through 0x9bf). |
| 0xfce  | <a href="#">PAGE39_LOCK0</a> | Lock configuration LSBs for page 39 (rows 0x9c0 through 0x9ff). |
| 0xfcf  | <a href="#">PAGE39_LOCK1</a> | Lock configuration MSBs for page 39 (rows 0x9c0 through 0x9ff). |
| 0xfd0  | <a href="#">PAGE40_LOCK0</a> | Lock configuration LSBs for page 40 (rows 0xa00 through 0xa3f). |
| 0xfd1  | <a href="#">PAGE40_LOCK1</a> | Lock configuration MSBs for page 40 (rows 0xa00 through 0xa3f). |
| 0xfd2  | <a href="#">PAGE41_LOCK0</a> | Lock configuration LSBs for page 41 (rows 0xa40 through 0xa7f). |
| 0xfd3  | <a href="#">PAGE41_LOCK1</a> | Lock configuration MSBs for page 41 (rows 0xa40 through 0xa7f). |
| 0xfd4  | <a href="#">PAGE42_LOCK0</a> | Lock configuration LSBs for page 42 (rows 0xa80 through 0xabf). |
| 0xfd5  | <a href="#">PAGE42_LOCK1</a> | Lock configuration MSBs for page 42 (rows 0xa80 through 0xabf). |
| 0xfd6  | <a href="#">PAGE43_LOCK0</a> | Lock configuration LSBs for page 43 (rows 0xac0 through 0xaff). |
| 0xfd7  | <a href="#">PAGE43_LOCK1</a> | Lock configuration MSBs for page 43 (rows 0xac0 through 0xaff). |
| 0xfd8  | <a href="#">PAGE44_LOCK0</a> | Lock configuration LSBs for page 44 (rows 0xb00 through 0xb3f). |
| 0xfd9  | <a href="#">PAGE44_LOCK1</a> | Lock configuration MSBs for page 44 (rows 0xb00 through 0xb3f). |
| 0xfda  | <a href="#">PAGE45_LOCK0</a> | Lock configuration LSBs for page 45 (rows 0xb40 through 0xb7f). |
| 0xfdb  | <a href="#">PAGE45_LOCK1</a> | Lock configuration MSBs for page 45 (rows 0xb40 through 0xb7f). |
| 0xfdc  | <a href="#">PAGE46_LOCK0</a> | Lock configuration LSBs for page 46 (rows 0xb80 through 0xbbf). |
| 0xfdd  | <a href="#">PAGE46_LOCK1</a> | Lock configuration MSBs for page 46 (rows 0xb80 through 0xbbf). |
| 0xfde  | <a href="#">PAGE47_LOCK0</a> | Lock configuration LSBs for page 47 (rows 0xbc0 through 0xbff). |
| 0xfdf  | <a href="#">PAGE47_LOCK1</a> | Lock configuration MSBs for page 47 (rows 0xbc0 through 0xbff). |
| 0xfe0  | <a href="#">PAGE48_LOCK0</a> | Lock configuration LSBs for page 48 (rows 0xc00 through 0xc3f). |
| 0xfe1  | <a href="#">PAGE48_LOCK1</a> | Lock configuration MSBs for page 48 (rows 0xc00 through 0xc3f). |
| 0xfe2  | <a href="#">PAGE49_LOCK0</a> | Lock configuration LSBs for page 49 (rows 0xc40 through 0xc7f). |

| Offset | Name                         | Info                                                            |
|--------|------------------------------|-----------------------------------------------------------------|
| 0xfe3  | <a href="#">PAGE49_LOCK1</a> | Lock configuration MSBs for page 49 (rows 0xc40 through 0xc7f). |
| 0xfe4  | <a href="#">PAGE50_LOCK0</a> | Lock configuration LSBs for page 50 (rows 0xc80 through 0xcbf). |
| 0xfe5  | <a href="#">PAGE50_LOCK1</a> | Lock configuration MSBs for page 50 (rows 0xc80 through 0xcbf). |
| 0xfe6  | <a href="#">PAGE51_LOCK0</a> | Lock configuration LSBs for page 51 (rows 0xcc0 through 0xcff). |
| 0xfe7  | <a href="#">PAGE51_LOCK1</a> | Lock configuration MSBs for page 51 (rows 0xcc0 through 0xcff). |
| 0xfe8  | <a href="#">PAGE52_LOCK0</a> | Lock configuration LSBs for page 52 (rows 0xd00 through 0xd3f). |
| 0xfe9  | <a href="#">PAGE52_LOCK1</a> | Lock configuration MSBs for page 52 (rows 0xd00 through 0xd3f). |
| 0xfea  | <a href="#">PAGE53_LOCK0</a> | Lock configuration LSBs for page 53 (rows 0xd40 through 0xd7f). |
| 0xfeb  | <a href="#">PAGE53_LOCK1</a> | Lock configuration MSBs for page 53 (rows 0xd40 through 0xd7f). |
| 0xfec  | <a href="#">PAGE54_LOCK0</a> | Lock configuration LSBs for page 54 (rows 0xd80 through 0xdbf). |
| 0xfed  | <a href="#">PAGE54_LOCK1</a> | Lock configuration MSBs for page 54 (rows 0xd80 through 0xdbf). |
| 0fee   | <a href="#">PAGE55_LOCK0</a> | Lock configuration LSBs for page 55 (rows 0xdc0 through 0xdff). |
| 0xfef  | <a href="#">PAGE55_LOCK1</a> | Lock configuration MSBs for page 55 (rows 0xdc0 through 0xdff). |
| 0xff0  | <a href="#">PAGE56_LOCK0</a> | Lock configuration LSBs for page 56 (rows 0xe00 through 0xe3f). |
| 0xff1  | <a href="#">PAGE56_LOCK1</a> | Lock configuration MSBs for page 56 (rows 0xe00 through 0xe3f). |
| 0xff2  | <a href="#">PAGE57_LOCK0</a> | Lock configuration LSBs for page 57 (rows 0xe40 through 0xe7f). |
| 0xff3  | <a href="#">PAGE57_LOCK1</a> | Lock configuration MSBs for page 57 (rows 0xe40 through 0xe7f). |
| 0xff4  | <a href="#">PAGE58_LOCK0</a> | Lock configuration LSBs for page 58 (rows 0xe80 through 0xebf). |
| 0xff5  | <a href="#">PAGE58_LOCK1</a> | Lock configuration MSBs for page 58 (rows 0xe80 through 0xebf). |
| 0xff6  | <a href="#">PAGE59_LOCK0</a> | Lock configuration LSBs for page 59 (rows 0xec0 through 0xeff). |
| 0xff7  | <a href="#">PAGE59_LOCK1</a> | Lock configuration MSBs for page 59 (rows 0xec0 through 0xeff). |
| 0xff8  | <a href="#">PAGE60_LOCK0</a> | Lock configuration LSBs for page 60 (rows 0xf00 through 0xf3f). |
| 0xff9  | <a href="#">PAGE60_LOCK1</a> | Lock configuration MSBs for page 60 (rows 0xf00 through 0xf3f). |

| Offset | Name         | Info                                                             |
|--------|--------------|------------------------------------------------------------------|
| 0xffa  | PAGE61_LOCK0 | Lock configuration LSBs for page 61 (rows 0xf40 through 0xf7f).  |
| 0xffb  | PAGE61_LOCK1 | Lock configuration MSBs for page 61 (rows 0xf40 through 0xf7f).  |
| 0xffc  | PAGE62_LOCK0 | Lock configuration LSBs for page 62 (rows 0xf80 through 0xfb7f). |
| 0xffd  | PAGE62_LOCK1 | Lock configuration MSBs for page 62 (rows 0xf80 through 0xfb7f). |
| 0xffe  | PAGE63_LOCK0 | Lock configuration LSBs for page 63 (rows 0xfc0 through 0xff7f). |
| 0xffff | PAGE63_LOCK1 | Lock configuration MSBs for page 63 (rows 0xfc0 through 0xff7f). |

## OTP\_DATA: CHIPID0 Register

Offset: 0x000

Table 1362. CHIPID0 Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Type | Reset |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -    | -     |
| 15:0  | <p>Bits 15:0 of public device ID. (ECC)</p> <p>The CHIPID0..3 rows contain a 64-bit random identifier for this chip, which can be read from the USB bootloader PICOBOOT interface or from the get_sys_info ROM API.</p> <p>The number of random bits makes the occurrence of twins exceedingly unlikely: for example, a fleet of a hundred million devices has a 99.97% probability of no twinned IDs. This is estimated to be lower than the occurrence of process errors in the assignment of sequential random IDs, and for practical purposes CHIPID may be treated as unique.</p> | RO   | -     |

## OTP\_DATA: CHIPID1 Register

Offset: 0x001

Table 1363. CHIPID1 Register

| Bits  | Description                          | Type | Reset |
|-------|--------------------------------------|------|-------|
| 31:16 | Reserved.                            | -    | -     |
| 15:0  | Bits 31:16 of public device ID (ECC) | RO   | -     |

## OTP\_DATA: CHIPID2 Register

Offset: 0x002

Table 1364. CHIPID2 Register

| Bits  | Description                          | Type | Reset |
|-------|--------------------------------------|------|-------|
| 31:16 | Reserved.                            | -    | -     |
| 15:0  | Bits 47:32 of public device ID (ECC) | RO   | -     |

## OTP\_DATA: CHIPID3 Register

Offset: 0x003

Table 1365. CHIPID3 Register

| Bits  | Description | Type | Reset |
|-------|-------------|------|-------|
| 31:16 | Reserved.   | -    | -     |

| Bits | Description                          | Type | Reset |
|------|--------------------------------------|------|-------|
| 15:0 | Bits 63:48 of public device ID (ECC) | RO   | -     |

## OTP\_DATA: RANDID0 Register

Offset: 0x004

Table 1366. RANDID0 Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Type | Reset |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -    | -     |
| 15:0  | <p>Bits 15:0 of private per-device random number (ECC)</p> <p>The RANDID0..7 rows form a 128-bit random number generated during device test.</p> <p>This ID is not exposed through the USB PICOB00T GET_INFO command or the ROM <code>get_sys_info()</code> API. However note that the USB PICOB00T OTP access point can read the entirety of page 0, so this value is not meaningfully private unless the USB PICOB00T interface is disabled via the DISABLE_BOOTSEL_USB_PICOB00T_IFC flag in BOOT_FLAGS0.</p> | RO   | -     |

## OTP\_DATA: RANDID1 Register

Offset: 0x005

Table 1367. RANDID1 Register

| Bits  | Description                                          | Type | Reset |
|-------|------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                            | -    | -     |
| 15:0  | Bits 31:16 of private per-device random number (ECC) | RO   | -     |

## OTP\_DATA: RANDID2 Register

Offset: 0x006

Table 1368. RANDID2 Register

| Bits  | Description                                          | Type | Reset |
|-------|------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                            | -    | -     |
| 15:0  | Bits 47:32 of private per-device random number (ECC) | RO   | -     |

## OTP\_DATA: RANDID3 Register

Offset: 0x007

Table 1369. RANDID3 Register

| Bits  | Description                                          | Type | Reset |
|-------|------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                            | -    | -     |
| 15:0  | Bits 63:48 of private per-device random number (ECC) | RO   | -     |

## OTP\_DATA: RANDID4 Register

Offset: 0x008

Table 1370. RANDID4 Register

| Bits  | Description                                          | Type | Reset |
|-------|------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                            | -    | -     |
| 15:0  | Bits 79:64 of private per-device random number (ECC) | RO   | -     |

**OTP\_DATA: RANDID5 Register**

Offset: 0x009

Table 1371. RANDID5 Register

| Bits  | Description                                          | Type | Reset |
|-------|------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                            | -    | -     |
| 15:0  | Bits 95:80 of private per-device random number (ECC) | RO   | -     |

**OTP\_DATA: RANDID6 Register**

Offset: 0x00a

Table 1372. RANDID6 Register

| Bits  | Description                                           | Type | Reset |
|-------|-------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                             | -    | -     |
| 15:0  | Bits 111:96 of private per-device random number (ECC) | RO   | -     |

**OTP\_DATA: RANDID7 Register**

Offset: 0x00b

Table 1373. RANDID7 Register

| Bits  | Description                                            | Type | Reset |
|-------|--------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                              | -    | -     |
| 15:0  | Bits 127:112 of private per-device random number (ECC) | RO   | -     |

**OTP\_DATA: ROSC\_CALIB Register**

Offset: 0x010

Table 1374. ROSC\_CALIB Register

| Bits  | Description                                                                                                                                                                              | Type | Reset |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                | -    | -     |
| 15:0  | Ring oscillator frequency in kHz, measured during manufacturing (ECC)<br><br>This is measured at 1.1 V, at room temperature, with the ROSC configuration registers in their reset state. | RO   | -     |

**OTP\_DATA: LPOSC\_CALIB Register**

Offset: 0x011

Table 1375.  
LPOSC\_CALIB Register

| Bits  | Description                                                                                                                                                                      | Type | Reset |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                        | -    | -     |
| 15:0  | Low-power oscillator frequency in Hz, measured during manufacturing (ECC)<br><br>This is measured at 1.1V, at room temperature, with the LPOSC trim register in its reset state. | RO   | -     |

## OTP\_DATA: NUM\_GPIOs Register

Offset: 0x018

Table 1376.  
NUM\_GPIOs Register

| Bits | Description                                                                                                     | Type | Reset |
|------|-----------------------------------------------------------------------------------------------------------------|------|-------|
| 31:8 | Reserved.                                                                                                       | -    | -     |
| 7:0  | The number of main user GPIOs (bank 0). Should read 48 in the QFN80 package, and 30 in the QFN60 package. (ECC) | RO   | -     |

## OTP\_DATA: INFO\_CRC0 Register

Offset: 0x036

Table 1377.  
INFO\_CRC0 Register

| Bits  | Description                                                                                                                                                  | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                    | -    | -     |
| 15:0  | Lower 16 bits of CRC32 of OTP addresses 0x00 through 0x6b (polynomial 0x4c11db7, input reflected, output reflected, seed all-ones, final XOR all-ones) (ECC) | RO   | -     |

## OTP\_DATA: INFO\_CRC1 Register

Offset: 0x037

Table 1378.  
INFO\_CRC1 Register

| Bits  | Description                                                     | Type | Reset |
|-------|-----------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                       | -    | -     |
| 15:0  | Upper 16 bits of CRC32 of OTP addresses 0x00 through 0x6b (ECC) | RO   | -     |

## OTP\_DATA: CRIT0 Register

Offset: 0x038

### Description

Page 0 critical boot flags (RBIT-8)

Table 1379. CRIT0 Register

| Bits | Description                                                            | Type | Reset |
|------|------------------------------------------------------------------------|------|-------|
| 31:2 | Reserved.                                                              | -    | -     |
| 1    | <b>RISCV_DISABLE</b> : Permanently disable RISC-V processors (Hazard3) | RO   | -     |
| 0    | <b>ARM_DISABLE</b> : Permanently disable ARM processors (Cortex-M33)   | RO   | -     |

## OTP\_DATA: CRIT0\_R1, CRIT0\_R2, ..., CRIT0\_R6, CRIT0\_R7 Registers

Offsets: 0x039, 0x03a, ..., 0x03e, 0x03f

Table 1380. CRIT0\_R1, CRIT0\_R2, ..., CRIT0\_R6, CRIT0\_R7 Registers

| Bits  | Description             | Type | Reset |
|-------|-------------------------|------|-------|
| 31:24 | Reserved.               | -    | -     |
| 23:0  | Redundant copy of CRIT0 | RO   | -     |

## OTP\_DATA: CRIT1 Register

Offset: 0x040

### Description

Page 1 critical boot flags (RBIT-8)

Table 1381. CRIT1 Register

| Bits | Description                                                                                                                                | Type | Reset |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:7 | Reserved.                                                                                                                                  | -    | -     |
| 6:5  | <b>GLITCH_DETECTOR_SENS</b> : Increase the sensitivity of the glitch detectors from their default.                                         | RO   | -     |
| 4    | <b>GLITCH_DETECTOR_ENABLE</b> : Arm the glitch detectors to reset the system if an abnormal clock/power event is observed.                 | RO   | -     |
| 3    | <b>BOOT_ARCH</b> : Set the default boot architecture, 0=ARM 1=RISC-V. Ignored if ARM_DISABLE, RISC_V_DISABLE or SECURE_BOOT_ENABLE is set. | RO   | -     |
| 2    | <b>DEBUG_DISABLE</b> : Disable all debug access                                                                                            | RO   | -     |
| 1    | <b>SECURE_DEBUG_DISABLE</b> : Disable Secure debug access                                                                                  | RO   | -     |
| 0    | <b>SECURE_BOOT_ENABLE</b> : Enable boot signature enforcement, and permanently disable the RISC-V cores.                                   | RO   | -     |

## OTP\_DATA: CRIT1\_R1, CRIT1\_R2, ..., CRIT1\_R6, CRIT1\_R7 Registers

Offsets: 0x041, 0x042, ..., 0x046, 0x047

Table 1382. CRIT1\_R1, CRIT1\_R2, ..., CRIT1\_R6, CRIT1\_R7 Registers

| Bits  | Description             | Type | Reset |
|-------|-------------------------|------|-------|
| 31:24 | Reserved.               | -    | -     |
| 23:0  | Redundant copy of CRIT1 | RO   | -     |

## OTP\_DATA: BOOT\_FLAGS0 Register

Offset: 0x048

### Description

Disable/Enable boot paths/features in the RP2350 mask ROM. Disables always supersede enables. Enables are provided where there are other configurations in OTP that must be valid. (RBIT-3)

Table 1383. BOOT\_FLAGS0 Register

| Bits  | Description                                                                                                                                                                                                                      | Type | Reset |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:22 | Reserved.                                                                                                                                                                                                                        | -    | -     |
| 21    | <b>DISABLE_SRAM_WINDOW_BOOT</b>                                                                                                                                                                                                  | RO   | -     |
| 20    | <b>DISABLE_XIP_ACCESS_ON_SRAM_ENTRY</b> : Disable all access to XIP after entering an SRAM binary.<br><br>Note that this will cause bootrom APIs that access XIP to fail, including APIs that interact with the partition table. | RO   | -     |
| 19    | <b>DISABLE_BOOTSEL_UART_BOOT</b>                                                                                                                                                                                                 | RO   | -     |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Type | Reset |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 18   | <b>DISABLE_BOOTSEL_USB_PICOBOOT_IFC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | RO   | -     |
| 17   | <b>DISABLE_BOOTSEL_USB_MSD_IFC</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | RO   | -     |
| 16   | <b>DISABLE_WATCHDOG_SCRATCH</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | RO   | -     |
| 15   | <b>DISABLE_POWER_SCRATCH</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | RO   | -     |
| 14   | <p><b>ENABLE_OTP_BOOT</b>: Enable OTP boot. A number of OTP rows specified by OTPBOOT_LEN will be loaded, starting from OTPBOOT_SRC, into the SRAM location specified by OTPBOOT_DST1 and OTPBOOT_DST0.</p> <p>The loaded program image is stored with ECC, 16 bits per row, and must contain a valid IMAGE_DEF. Do not set this bit without first programming an image into OTP and configuring OTPBOOT_LEN, OTPBOOT_SRC, OTPBOOT_DST0 and OTPBOOT_DST1.</p> <p>Note that OTPBOOT_LEN and OTPBOOT_SRC must be even numbers of OTP rows. Equivalently, the image must be a multiple of 32 bits in size, and must start at a 32-bit-aligned address in the ECC read data address window.</p> | RO   | -     |
| 13   | <b>DISABLE_OTP_BOOT</b> : Takes precedence over ENABLE_OTP_BOOT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RO   | -     |
| 12   | <b>DISABLE_FLASH_BOOT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | RO   | -     |
| 11   | <b>ROLLBACK_REQUIRED</b> : Require binaries to have a rollback version. Set automatically the first time a binary with a rollback version is booted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | RO   | -     |
| 10   | <b>HASHED_PARTITION_TABLE</b> : Require a partition table to be hashed (if not signed)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | RO   | -     |
| 9    | <b>SECURE_PARTITION_TABLE</b> : Require a partition table to be signed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | RO   | -     |
| 8    | <b>DISABLE_AUTO_SWITCH_ARCH</b> : Disable auto-switch of CPU architecture on boot when the (only) binary to be booted is for the other Arm/RISC-V architecture and both architectures are enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | RO   | -     |
| 7    | <b>SINGLE_FLASH_BINARY</b> : Restrict flash boot path to use of a single binary at the start of flash                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RO   | -     |
| 6    | <p><b>OVERRIDE_FLASH_PARTITION_SLOT_SIZE</b>: Override the limit for default flash metadata scanning.</p> <p>The value is specified in FLASH_PARTITION_SLOT_SIZE. Make sure FLASH_PARTITION_SLOT_SIZE is valid before setting this bit</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | RO   | -     |
| 5    | <b>FLASH_DEVINFO_ENABLE</b> : Mark FLASH_DEVINFO as containing valid, ECC'd data which describes external flash devices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RO   | -     |
| 4    | <b>FAST_SIGCHECK_ROSC_DIV</b> : Enable quartering of ROSC divisor during signature check, to reduce secure boot time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | RO   | -     |
| 3    | <p><b>FLASH_IO_VOLTAGE_1V8</b>: If 1, configure the QSPI pads for 1.8 V operation when accessing flash for the first time from the bootrom, using the VOLTAGE_SELECT register for the QSPI pads bank. This slightly improves the input timing of the pads at low voltages, but does not affect their output characteristics.</p> <p>If 0, leave VOLTAGE_SELECT in its reset state (suitable for operation at and above 2.5 V)</p>                                                                                                                                                                                                                                                           | RO   | -     |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                      | Type | Reset |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 2    | <b>ENABLE_BOOTSEL_NON_DEFAULT_PLL_XOSC_CFG</b> : Enable loading of the non-default XOSC and PLL configuration before entering BOOTSEL mode.<br><br>Ensure that BOOTSEL_XOSC_CFG and BOOTSEL_PLL_CFG are correctly programmed before setting this bit.<br><br>If this bit is set, user software may use the contents of BOOTSEL_PLL_CFG to calculate the expected XOSC frequency based on the fixed USB boot frequency of 48 MHz. | RO   | -     |
| 1    | <b>ENABLE_BOOTSEL_LED</b> : Enable bootloader activity LED. If set, bootsel_led_cfg is assumed to be valid                                                                                                                                                                                                                                                                                                                       | RO   | -     |
| 0    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                        | -    | -     |

## OTP\_DATA: BOOT\_FLAGS0\_R1, BOOT\_FLAGS0\_R2 Registers

**Offsets:** 0x049, 0x04a

Table 1384.  
BOOT\_FLAGS0\_R1,  
BOOT\_FLAGS0\_R2  
Registers

| Bits  | Description                   | Type | Reset |
|-------|-------------------------------|------|-------|
| 31:24 | Reserved.                     | -    | -     |
| 23:0  | Redundant copy of BOOT_FLAGS0 | RO   | -     |

## OTP\_DATA: BOOT\_FLAGS1 Register

**Offset:** 0x04b

### Description

Disable/Enable boot paths/features in the RP2350 mask ROM. Disables always supersede enables. Enables are provided where there are other configurations in OTP that must be valid. (RBIT-3)

Table 1385.  
BOOT\_FLAGS1  
Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Type | Reset |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:20 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -    | -     |
| 19    | <b>DOUBLE_TAP</b> : Enable entering BOOTSEL mode via double-tap of the RUN/RSTn pin. Adds a significant delay to boot time, as configured by DOUBLE_TAP_DELAY.<br><br>This functions by waiting at startup (i.e. following a reset) to see if a second reset is applied soon afterward. The second reset is detected by the bootrom with help of the POWMAN_CHIP_RESET_DOUBLE_TAP flag, which is not reset by the external reset pin, and the bootrom enters BOOTSEL mode (NSBOOT) to await further instruction over USB or UART. | RO   | -     |
| 18:16 | <b>DOUBLE_TAP_DELAY</b> : Adjust how long to wait for a second reset when double tap BOOTSEL mode is enabled via DOUBLE_TAP. The minimum is 50 milliseconds, and each unit of this field adds an additional 50 milliseconds.<br><br>For example, settings this field to its maximum value of 7 will cause the chip to wait for 400 milliseconds at boot to check for a second reset which requests entry to BOOTSEL mode.<br><br>200 milliseconds (DOUBLE_TAP_DELAY=3) is a good intermediate value.                              | RO   | -     |
| 15:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -    | -     |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Type | Reset |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 11:8 | <p><b>KEY_INVALID:</b> Mark a boot key as invalid, or prevent it from ever becoming valid. The bootrom will ignore any boot key marked as invalid during secure boot signature checks.</p> <p>Each bit in this field corresponds to one of the four 256-bit boot key hashes that may be stored in page 2 of the OTP.</p> <p>When provisioning boot keys, it's recommended to mark any boot key slots you don't intend to use as KEY_INVALID, so that spurious keys can not be installed at a later time.</p>                                                                                                                                                                                                                                                                                                                                                                        | RO   | -     |
| 7:4  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -    | -     |
| 3:0  | <p><b>KEY_VALID:</b> Mark each of the possible boot keys as valid. The bootrom will check signatures against all valid boot keys, and ignore invalid boot keys.</p> <p>Each bit in this field corresponds to one of the four 256-bit boot key hashes that may be stored in page 2 of the OTP.</p> <p>A KEY_VALID bit is ignored if the corresponding KEY_INVALID bit is set. Boot keys are considered valid only when KEY_VALID is set and KEY_INVALID is clear.</p> <p>Do not mark a boot key as KEY_VALID if it does not contain a valid SHA-256 hash of your secp256k1 public key. Verify keys after programming, before setting the KEY_VALID bits – a boot key with uncorrectable ECC faults will render your device unbootable if secure boot is enabled.</p> <p>Do not enable secure boot without first installing a valid key. This will render your device unbootable.</p> | RO   | -     |

## OTP\_DATA: BOOT\_FLAGS1\_R1, BOOT\_FLAGS1\_R2 Registers

Offsets: 0x04c, 0x04d

Table 1386.  
BOOT\_FLAGS1\_R1,  
BOOT\_FLAGS1\_R2  
Registers

| Bits  | Description                   | Type | Reset |
|-------|-------------------------------|------|-------|
| 31:24 | Reserved.                     | -    | -     |
| 23:0  | Redundant copy of BOOT_FLAGS1 | RO   | -     |

## OTP\_DATA: DEFAULT\_BOOT\_VERSION0 Register

Offset: 0x04e

Table 1387.  
DEFAULT\_BOOT\_VERSION0  
Register

| Bits  | Description                                                  | Type | Reset |
|-------|--------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                    | -    | -     |
| 23:0  | Default boot version thermometer counter, bits 23:0 (RBIT-3) | RO   | -     |

## OTP\_DATA: DEFAULT\_BOOT\_VERSION0\_R1, DEFAULT\_BOOT\_VERSION0\_R2 Registers

Offsets: 0x04f, 0x050

Table 1388.  
DEFAULT\_BOOT\_VERS  
ION0\_R1,  
DEFAULT\_BOOT\_VERS  
ION0\_R2 Registers

| Bits  | Description                             | Type | Reset |
|-------|-----------------------------------------|------|-------|
| 31:24 | Reserved.                               | -    | -     |
| 23:0  | Redundant copy of DEFAULT_BOOT_VERSION0 | RO   | -     |

## OTP\_DATA: DEFAULT\_BOOT\_VERSION1 Register

Offset: 0x051

Table 1389.  
DEFAULT\_BOOT\_VERS  
ION1 Register

| Bits  | Description                                                   | Type | Reset |
|-------|---------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                     | -    | -     |
| 23:0  | Default boot version thermometer counter, bits 47:24 (RBIT-3) | RO   | -     |

## OTP\_DATA: DEFAULT\_BOOT\_VERSION1\_R1, DEFAULT\_BOOT\_VERSION1\_R2 Registers

Offsets: 0x052, 0x053

Table 1390.  
DEFAULT\_BOOT\_VERS  
ION1\_R1,  
DEFAULT\_BOOT\_VERS  
ION1\_R2 Registers

| Bits  | Description                             | Type | Reset |
|-------|-----------------------------------------|------|-------|
| 31:24 | Reserved.                               | -    | -     |
| 23:0  | Redundant copy of DEFAULT_BOOT_VERSION1 | RO   | -     |

## OTP\_DATA: FLASH\_DEVINFO Register

Offset: 0x054

### Description

Stores information about external flash device(s). (ECC)

Assumed to be valid if BOOT\_FLAGS0\_FLASH\_DEVINFO\_ENABLE is set.

Table 1391.  
FLASH\_DEVINFO  
Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Type | Reset |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -    | -     |
| 15:12 | <p><b>CS1_SIZE:</b> The size of the flash/PSRAM device on chip select 1 (addressable at 0x11000000 through 0x11ffffff).</p> <p>A value of zero is decoded as a size of zero (no device). Nonzero values are decoded as 4kiB &lt;&lt; CS1_SIZE. For example, four megabytes is encoded with a CS1_SIZE value of 10, and 16 megabytes is encoded with a CS1_SIZE value of 12.</p> <p>When BOOT_FLAGS0_FLASH_DEVINFO_ENABLE is not set, a default of zero is used.</p> | RO   | -     |
|       | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |       |
|       | 0x0 → NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
|       | 0x1 → 8K                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |       |
|       | 0x2 → 16K                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|       | 0x3 → 32K                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|       | 0x4 → 64K                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|       | 0x5 → 128K                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Type | Reset |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
|      | 0x6 → 256K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |
|      | 0x7 → 512K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |
|      | 0x8 → 1M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0x9 → 2M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0xa → 4M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0xb → 8M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0xc → 16M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
| 11:8 | <p><b>CS0_SIZE:</b> The size of the flash/PSRAM device on chip select 0 (addressable at 0x10000000 through 0x10ffffff).</p> <p>A value of zero is decoded as a size of zero (no device). Nonzero values are decoded as 4kiB &lt;&lt; CS0_SIZE. For example, four megabytes is encoded with a CS0_SIZE value of 10, and 16 megabytes is encoded with a CS0_SIZE value of 12.</p> <p>When BOOT_FLAGS0_FLASH_DEVINFO_ENABLE is not set, a default of 12 (16 MiB) is used.</p>                                                         | RO   | -     |
|      | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |       |
|      | 0x0 → NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |
|      | 0x1 → 8K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0x2 → 16K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
|      | 0x3 → 32K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
|      | 0x4 → 64K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
|      | 0x5 → 128K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |
|      | 0x6 → 256K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |
|      | 0x7 → 512K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |       |
|      | 0x8 → 1M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0x9 → 2M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0xa → 4M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0xb → 8M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0xc → 16M                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
| 7    | <p><b>D8H_ERASE_SUPPORTED:</b> If true, all attached devices are assumed to support (or ignore, in the case of PSRAM) a block erase command with a command prefix of D8h, an erase size of 64 kiB, and a 24-bit address. Almost all 25-series flash devices support this command.</p> <p>If set, the bootrom will use the D8h erase command where it is able, to accelerate bulk erase operations. This makes flash programming faster.</p> <p>When BOOT_FLAGS0_FLASH_DEVINFO_ENABLE is not set, this field defaults to false.</p> | RO   | -     |
| 6    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | -    | -     |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Type | Reset |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 5:0  | <p><b>CS1_GPIO:</b> Indicate a GPIO number to be used for the secondary flash chip select (CS1), which selects the external QSPI device mapped at system addresses 0x11000000 through 0x11ffffff. There is no such configuration for CS0, as the primary chip select has a dedicated pin.</p> <p>On RP2350 the permissible GPIO numbers are 0, 8, 19 and 47.</p> <p>Ignored if CS1_size is zero. If CS1_SIZE is nonzero, the bootrom will automatically configure this GPIO as a second chip select upon entering the flash boot path, or entering any other path that may use the QSPI flash interface, such as BOOTSEL mode (nsboot).</p> | RO   | -     |

## OTP\_DATA: FLASH\_PARTITION\_SLOT\_SIZE Register

Offset: 0x055

Table 1392.  
FLASH\_PARTITION\_SLOT\_SIZE Register

| Bits  | Description                                                                                                                                                                                                    | Type | Reset |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                      | -    | -     |
| 15:0  | Gap between partition table slot 0 and slot 1 at the start of flash (the default size is 4096 bytes) (ECC) Enabled by the OVERRIDE_FLASH_PARTITION_SLOT_SIZE bit in BOOT_FLAGS, the size is 4096 * (value + 1) | RO   | -     |

## OTP\_DATA: BOOTSEL\_LED\_CFG Register

Offset: 0x056

### Description

Pin configuration for LED status, used by USB bootloader. (ECC)  
Must be valid if BOOT\_FLAGS0\_ENABLE\_BOOTSEL\_LED is set.

Table 1393.  
BOOTSEL\_LED\_CFG Register

| Bits | Description                                                  | Type | Reset |
|------|--------------------------------------------------------------|------|-------|
| 31:9 | Reserved.                                                    | -    | -     |
| 8    | <b>ACTIVELOW:</b> LED is active-low. (Default: active-high.) | RO   | -     |
| 7:6  | Reserved.                                                    | -    | -     |
| 5:0  | <b>PIN:</b> GPIO index to use for bootloader activity LED.   | RO   | -     |

## OTP\_DATA: BOOTSEL\_PLL\_CFG Register

Offset: 0x057

### Description

Optional PLL configuration for BOOTSEL mode. (ECC)

This should be configured to produce an exact 48 MHz based on the crystal oscillator frequency. User mode software may also use this value to calculate the expected crystal frequency based on an assumed 48 MHz PLL output.

If no configuration is given, the crystal is assumed to be 12 MHz.

The PLL frequency can be calculated as:

$$\text{PLL out} = (\text{XOSC frequency} / (\text{REFDIV} + 1)) \times \text{FBDIV} / (\text{POSTDIV1} \times \text{POSTDIV2})$$

Conversely the crystal frequency can be calculated as:

$$\text{XOSC frequency} = 48 \text{ MHz} \times (\text{REFDIV} + 1) \times (\text{POSTDIV1} \times \text{POSTDIV2}) / \text{FBDIV}$$

(Note the +1 on REFDIV is because the value stored in this OTP location is the actual divisor value minus one.)

Used if and only if ENABLE\_BOOTSEL\_NON\_DEFAULT\_PLL\_XOSC\_CFG is set in BOOT\_FLAGS0. That bit should be set only after this row and BOOTSEL\_XOSC\_CFG are both correctly programmed.

Table 1394.  
BOOTSEL\_PLL\_CFG  
Register

| Bits  | Description                                                                                                                                                                                                   | Type | Reset |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                     | -    | -     |
| 15    | <b>REFDIV</b> : PLL reference divisor, minus one.<br><br>Programming a value of 0 means a reference divisor of 1. Programming a value of 1 means a reference divisor of 2 (for exceptionally fast XIN inputs) | RO   | -     |
| 14:12 | <b>POSTDIV2</b> : PLL post-divide 2 divisor, in the range 1..7 inclusive.                                                                                                                                     | RO   | -     |
| 11:9  | <b>POSTDIV1</b> : PLL post-divide 1 divisor, in the range 1..7 inclusive.                                                                                                                                     | RO   | -     |
| 8:0   | <b>FBDIV</b> : PLL feedback divisor, in the range 16..320 inclusive.                                                                                                                                          | RO   | -     |

## OTP\_DATA: BOOTSEL\_XOSC\_CFG Register

**Offset:** 0x058

### Description

Non-default crystal oscillator configuration for the USB bootloader. (ECC)

These values may also be used by user code configuring the crystal oscillator.

Used if and only if ENABLE\_BOOTSEL\_NON\_DEFAULT\_PLL\_XOSC\_CFG is set in BOOT\_FLAGS0. That bit should be set only after this row and BOOTSEL\_PLL\_CFG are both correctly programmed.

Table 1395.  
BOOTSEL\_XOSC\_CFG  
Register

| Bits  | Description                                                | Type | Reset |
|-------|------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                  | -    | -     |
| 15:14 | <b>RANGE</b> : Value of the XOSC_CTRL_FREQ_RANGE register. | RO   | -     |
|       | Enumerated values:                                         |      |       |
|       | 0x0 → 1_15MHZ                                              |      |       |
|       | 0x1 → 10_30MHZ                                             |      |       |
|       | 0x2 → 25_60MHZ                                             |      |       |
|       | 0x3 → 40_100MHZ                                            |      |       |
| 13:0  | <b>STARTUP</b> : Value of the XOSC_STARTUP register        | RO   | -     |

## OTP\_DATA: USB\_BOOT\_FLAGS Register

**Offset:** 0x059

### Description

USB boot specific feature flags (RBIT-3)

Table 1396.  
USB\_BOOT\_FLAGS  
Register

| Bits  | Description                                                                                                                    | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                                                                                      | -    | -     |
| 23    | <b>DP_DM_SWAP</b> : Swap DM/DP during USB boot, to support board layouts with mirrored USB routing (deliberate or accidental). | RO   | -     |

| Bits  | Description                                                                                                                                                  | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 22    | <b>WHITE_LABEL_ADDR_VALID</b> : valid flag for INFO_UF2_TXT_BOARD_ID_STRDEF entry of the USB_WHITE_LABEL struct (index 15)                                   | RO   | -     |
| 21:16 | Reserved.                                                                                                                                                    | -    | -     |
| 15    | <b>WL_INFO_UF2_TXT_BOARD_ID_STRDEF_VALID</b> : valid flag for the USB_WHITE_LABEL_ADDR field                                                                 | RO   | -     |
| 14    | <b>WL_INFO_UF2_TXT_MODEL_STRDEF_VALID</b> : valid flag for INFO_UF2_TXT_MODEL_STRDEF entry of the USB_WHITE_LABEL struct (index 14)                          | RO   | -     |
| 13    | <b>WL_INDEX_HTM_REDIRECT_NAME_STRDEF_VALID</b> : valid flag for INDEX_HTM_REDIRECT_NAME_STRDEF entry of the USB_WHITE_LABEL struct (index 13)                | RO   | -     |
| 12    | <b>WL_INDEX_HTM_REDIRECT_URL_STRDEF_VALID</b> : valid flag for INDEX_HTM_REDIRECT_URL_STRDEF entry of the USB_WHITE_LABEL struct (index 12)                  | RO   | -     |
| 11    | <b>WL_SCSI_INQUIRY_VERSION_STRDEF_VALID</b> : valid flag for SCSI_INQUIRY_VERSION_STRDEF entry of the USB_WHITE_LABEL struct (index 11)                      | RO   | -     |
| 10    | <b>WL_SCSI_INQUIRY_PRODUCT_STRDEF_VALID</b> : valid flag for SCSI_INQUIRY_PRODUCT_STRDEF entry of the USB_WHITE_LABEL struct (index 10)                      | RO   | -     |
| 9     | <b>WL_SCSI_INQUIRY_VENDOR_STRDEF_VALID</b> : valid flag for SCSI_INQUIRY_VENDOR_STRDEF entry of the USB_WHITE_LABEL struct (index 9)                         | RO   | -     |
| 8     | <b>WL_VOLUME_LABEL_STRDEF_VALID</b> : valid flag for VOLUME_LABEL_STRDEF entry of the USB_WHITE_LABEL struct (index 8)                                       | RO   | -     |
| 7     | <b>WL_USB_CONFIG_ATTRIBUTES_MAX_POWER_VALUES_VALID</b> : valid flag for USB_CONFIG_ATTRIBUTES_MAX_POWER_VALUES entry of the USB_WHITE_LABEL struct (index 7) | RO   | -     |
| 6     | <b>WL_USB_DEVICE_SERIAL_NUMBER_STRDEF_VALID</b> : valid flag for USB_DEVICE_SERIAL_NUMBER_STRDEF entry of the USB_WHITE_LABEL struct (index 6)               | RO   | -     |
| 5     | <b>WL_USB_DEVICE_PRODUCT_STRDEF_VALID</b> : valid flag for USB_DEVICE_PRODUCT_STRDEF entry of the USB_WHITE_LABEL struct (index 5)                           | RO   | -     |
| 4     | <b>WL_USB_DEVICE_MANUFACTURER_STRDEF_VALID</b> : valid flag for USB_DEVICE_MANUFACTURER_STRDEF entry of the USB_WHITE_LABEL struct (index 4)                 | RO   | -     |
| 3     | <b>WL_USB_DEVICE_LANG_ID_VALUE_VALID</b> : valid flag for USB_DEVICE_LANG_ID_VALUE entry of the USB_WHITE_LABEL struct (index 3)                             | RO   | -     |
| 2     | <b>WL_USB_DEVICE_SERIAL_NUMBER_VALUE_VALID</b> : valid flag for USB_DEVICE_BCD_DEVICEVALUE entry of the USB_WHITE_LABEL struct (index 2)                     | RO   | -     |
| 1     | <b>WL_USB_DEVICE_PID_VALUE_VALID</b> : valid flag for USB_DEVICE_PID_VALUE entry of the USB_WHITE_LABEL struct (index 1)                                     | RO   | -     |

| Bits | Description                                                                                                              | Type | Reset |
|------|--------------------------------------------------------------------------------------------------------------------------|------|-------|
| 0    | <b>WL_USB_DEVICE_VID_VALUE_VALID</b> : valid flag for USB_DEVICE_VID_VALUE entry of the USB_WHITE_LABEL struct (index 0) | RO   | -     |

## OTP\_DATA: USB\_BOOT\_FLAGS\_R1, USB\_BOOT\_FLAGS\_R2 Registers

Offsets: 0x05a, 0x05b

Table 1397.  
USB\_BOOT\_FLAGS\_R1,  
USB\_BOOT\_FLAGS\_R2  
Registers

| Bits  | Description                      | Type | Reset |
|-------|----------------------------------|------|-------|
| 31:24 | Reserved.                        | -    | -     |
| 23:0  | Redundant copy of USB_BOOT_FLAGS | RO   | -     |

## OTP\_DATA: USB\_WHITE\_LABEL\_ADDR Register

Offset: 0x05c

Table 1398.  
USB\_WHITE\_LABEL\_A  
DDR Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Type | Reset |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -    | -     |
| 15:0  | <p>Row index of the USB_WHITE_LABEL structure within OTP (ECC)</p> <p>The table has 16 rows, each of which are also ECC and marked valid by the corresponding valid bit in USB_BOOT_FLAGS (ECC).</p> <p>The entries are either _VALUES where the 16 bit value is used as is, or _STRDEFs which acts as a pointers to a string value.</p> <p>The value stored in a _STRDEF is two separate bytes: The low seven bits of the first (LSB) byte indicates the number of characters in the string, and the top bit of the first (LSB) byte if set to indicate that each character in the string is two bytes (Unicode) versus one byte if unset. The second (MSB) byte represents the location of the string data, and is encoded as the number of rows from this USB_WHITE_LABEL_ADDR; i.e. the row of the start of the string is USB_WHITE_LABEL_ADDR value + msb_byte.</p> <p>In each case, the corresponding valid bit enables replacing the default value for the corresponding item provided by the boot rom.</p> <p>Note that Unicode _STRDEFs are only supported for USB_DEVICE_PRODUCT_STRDEF, USB_DEVICE_SERIAL_NUMBER_STRDEF and USB_DEVICE_MANUFACTURER_STRDEF. Unicode values will be ignored if specified for other fields, and non-unicode values for these three items will be converted to Unicode characters by setting the upper 8 bits to zero.</p> <p>Note that if the USB_WHITE_LABEL structure or the corresponding strings are not readable by BOOTSEL mode based on OTP permissions, or if alignment requirements are not met, then the corresponding default values are used.</p> <p>The index values indicate where each field is located (row USB_WHITE_LABEL_ADDR value + index):</p> | RO   | -     |
|       | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |       |
|       | 0x0000 → INDEX_USB_DEVICE_VID_VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|       | 0x0001 → INDEX_USB_DEVICE_PID_VALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |

| Bits | Description                                           | Type | Reset |
|------|-------------------------------------------------------|------|-------|
|      | 0x0002 → INDEX_USB_DEVICE_BCD_DEVICE_VALUE            |      |       |
|      | 0x0003 → INDEX_USB_DEVICE_LANG_ID_VALUE               |      |       |
|      | 0x0004 → INDEX_USB_DEVICE_MANUFACTURER_STRDEF         |      |       |
|      | 0x0005 → INDEX_USB_DEVICE_PRODUCT_STRDEF              |      |       |
|      | 0x0006 → INDEX_USB_DEVICE_SERIAL_NUMBER_STRDEF        |      |       |
|      | 0x0007 → INDEX_USB_CONFIG_ATTRIBUTES_MAX_POWER_VALUES |      |       |
|      | 0x0008 → INDEX_VOLUME_LABEL_STRDEF                    |      |       |
|      | 0x0009 → INDEX_SCSI_INQUIRY_VENDOR_STRDEF             |      |       |
|      | 0x000a → INDEX_SCSI_INQUIRY_PRODUCT_STRDEF            |      |       |
|      | 0x000b → INDEX_SCSI_INQUIRY_VERSION_STRDEF            |      |       |
|      | 0x000c → INDEX_INDEX_HTM_REDIRECT_URL_STRDEF          |      |       |
|      | 0x000d → INDEX_INDEX_HTM_REDIRECT_NAME_STRDEF         |      |       |
|      | 0x000e → INDEX_INFO_UF2_TXT_MODEL_STRDEF              |      |       |
|      | 0x000f → INDEX_INFO_UF2_TXT_BOARD_ID_STRDEF           |      |       |

## OTP\_DATA: OTPBOOT\_SRC Register

Offset: 0x05e

Table 1399.  
OTPBOOT\_SRC  
Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                      | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                        | -    | -     |
| 15:0  | <p>OTP start row for the OTP boot image. (ECC)</p> <p>If OTP boot is enabled, the bootrom will load from this location into SRAM and then directly enter the loaded image. Note that the image must be signed if SECURE_BOOT_ENABLE is set. The image itself is assumed to be ECC-protected.</p> <p>This must be an even number. Equivalently, the OTP boot image must start at a word-aligned location in the ECC read data address window.</p> | RO   | -     |

## OTP\_DATA: OTPBOOT\_LEN Register

Offset: 0x05f

Table 1400.  
OTPBOOT\_LEN  
Register

| Bits  | Description                                                                                                                                       | Type | Reset |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                         | -    | -     |
| 15:0  | <p>Length in rows of the OTP boot image. (ECC)</p> <p>OTPBOOT_LEN must be even. The total image size must be a multiple of 4 bytes (32 bits).</p> | RO   | -     |

## OTP\_DATA: OTPBOOT\_DST0 Register

Offset: 0x060

Table 1401.  
OTPBOOT\_DST0  
Register

| Bits  | Description                                                                                                                                                                                            | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                              | -    | -     |
| 15:0  | Bits 15:0 of the OTP boot image load destination (and entry point). (ECC)<br><br>This must be a location in main SRAM (main SRAM is addresses 0x20000000 through 0x20082000) and must be word-aligned. | RO   | -     |

## OTP\_DATA: OTPBOOT\_DST1 Register

Offset: 0x061

Table 1402.  
OTPBOOT\_DST1  
Register

| Bits  | Description                                                                                                                                                                                             | Type | Reset |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                                                                                                                                                                               | -    | -     |
| 15:0  | Bits 31:16 of the OTP boot image load destination (and entry point). (ECC)<br><br>This must be a location in main SRAM (main SRAM is addresses 0x20000000 through 0x20082000) and must be word-aligned. | RO   | -     |

## OTP\_DATA: BOOTKEY0\_0, BOOTKEY0\_1, ..., BOOTKEY3\_14, BOOTKEY3\_15 Registers

Offsets: 0x080, 0x081, ..., 0x0be, 0x0bf

Table 1403.  
BOOTKEY0\_0,  
BOOTKEY0\_1, ...,  
BOOTKEY3\_14,  
BOOTKEY3\_15  
Registers

| Bits  | Description                                             | Type | Reset |
|-------|---------------------------------------------------------|------|-------|
| 31:16 | Reserved.                                               | -    | -     |
| 15:0  | Bits $N + 15 : N$ of SHA-256 hash of boot key $K$ (ECC) | RO   | -     |

## OTP\_DATA: KEY1\_0, KEY2\_0, ..., KEY5\_0, KEY6\_0 Registers

Offsets: 0xf48, 0xf50, ..., 0xf68, 0xf70

Table 1404. KEY1\_0,  
KEY2\_0, ..., KEY5\_0,  
KEY6\_0 Registers

| Bits  | Description                           | Type | Reset |
|-------|---------------------------------------|------|-------|
| 31:16 | Reserved.                             | -    | -     |
| 15:0  | Bits 15:0 of OTP access key $n$ (ECC) | RO   | -     |

## OTP\_DATA: KEY1\_1, KEY2\_1, ..., KEY5\_1, KEY6\_1 Registers

Offsets: 0xf49, 0xf51, ..., 0xf69, 0xf71

Table 1405. KEY1\_1,  
KEY2\_1, ..., KEY5\_1,  
KEY6\_1 Registers

| Bits  | Description                            | Type | Reset |
|-------|----------------------------------------|------|-------|
| 31:16 | Reserved.                              | -    | -     |
| 15:0  | Bits 31:16 of OTP access key $n$ (ECC) | RO   | -     |

## OTP\_DATA: KEY1\_2, KEY2\_2, ..., KEY5\_2, KEY6\_2 Registers

Offsets: 0xf4a, 0xf52, ..., 0xf6a, 0xf72

Table 1406. KEY1\_2, KEY2\_2, ..., KEY5\_2, KEY6\_2 Registers

| Bits  | Description                            | Type | Reset |
|-------|----------------------------------------|------|-------|
| 31:16 | Reserved.                              | -    | -     |
| 15:0  | Bits 47:32 of OTP access key $n$ (ECC) | RO   | -     |

**OTP\_DATA: KEY1\_3, KEY2\_3, ..., KEY5\_3, KEY6\_3 Registers****Offsets:** 0xf4b, 0xf53, ..., 0xf6b, 0xf73

Table 1407. KEY1\_3, KEY2\_3, ..., KEY5\_3, KEY6\_3 Registers

| Bits  | Description                            | Type | Reset |
|-------|----------------------------------------|------|-------|
| 31:16 | Reserved.                              | -    | -     |
| 15:0  | Bits 63:48 of OTP access key $n$ (ECC) | RO   | -     |

**OTP\_DATA: KEY1\_4, KEY2\_4, ..., KEY5\_4, KEY6\_4 Registers****Offsets:** 0xf4c, 0xf54, ..., 0xf6c, 0xf74

Table 1408. KEY1\_4, KEY2\_4, ..., KEY5\_4, KEY6\_4 Registers

| Bits  | Description                            | Type | Reset |
|-------|----------------------------------------|------|-------|
| 31:16 | Reserved.                              | -    | -     |
| 15:0  | Bits 79:64 of OTP access key $n$ (ECC) | RO   | -     |

**OTP\_DATA: KEY1\_5, KEY2\_5, ..., KEY5\_5, KEY6\_5 Registers****Offsets:** 0xf4d, 0xf55, ..., 0xf6d, 0xf75

Table 1409. KEY1\_5, KEY2\_5, ..., KEY5\_5, KEY6\_5 Registers

| Bits  | Description                            | Type | Reset |
|-------|----------------------------------------|------|-------|
| 31:16 | Reserved.                              | -    | -     |
| 15:0  | Bits 95:80 of OTP access key $n$ (ECC) | RO   | -     |

**OTP\_DATA: KEY1\_6, KEY2\_6, ..., KEY5\_6, KEY6\_6 Registers****Offsets:** 0xf4e, 0xf56, ..., 0xf6e, 0xf76

Table 1410. KEY1\_6, KEY2\_6, ..., KEY5\_6, KEY6\_6 Registers

| Bits  | Description                             | Type | Reset |
|-------|-----------------------------------------|------|-------|
| 31:16 | Reserved.                               | -    | -     |
| 15:0  | Bits 111:96 of OTP access key $n$ (ECC) | RO   | -     |

**OTP\_DATA: KEY1\_7, KEY2\_7, ..., KEY5\_7, KEY6\_7 Registers****Offsets:** 0xf4f, 0xf57, ..., 0xf6f, 0xf77

Table 1411. KEY1\_7, KEY2\_7, ..., KEY5\_7, KEY6\_7 Registers

| Bits  | Description                              | Type | Reset |
|-------|------------------------------------------|------|-------|
| 31:16 | Reserved.                                | -    | -     |
| 15:0  | Bits 127:112 of OTP access key $n$ (ECC) | RO   | -     |

**OTP\_DATA: KEY1\_VALID Register****Offset:** 0xf79**Description**

Valid flag for key 1. Once the valid flag is set, the key can no longer be read or written, and becomes a valid fixed key for protecting OTP pages.

Table 1412.  
KEY1\_VALID Register

| Bits  | Description                                                         | Type | Reset |
|-------|---------------------------------------------------------------------|------|-------|
| 31:17 | Reserved.                                                           | -    | -     |
| 16    | <b>VALID_R2</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 15:9  | Reserved.                                                           | -    | -     |
| 8     | <b>VALID_R1</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 7:1   | Reserved.                                                           | -    | -     |
| 0     | <b>VALID</b>                                                        | RO   | -     |

## OTP\_DATA: KEY2\_VALID Register

**Offset:** 0xf7a

### Description

Valid flag for key 2. Once the valid flag is set, the key can no longer be read or written, and becomes a valid fixed key for protecting OTP pages.

Table 1413.  
KEY2\_VALID Register

| Bits  | Description                                                         | Type | Reset |
|-------|---------------------------------------------------------------------|------|-------|
| 31:17 | Reserved.                                                           | -    | -     |
| 16    | <b>VALID_R2</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 15:9  | Reserved.                                                           | -    | -     |
| 8     | <b>VALID_R1</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 7:1   | Reserved.                                                           | -    | -     |
| 0     | <b>VALID</b>                                                        | RO   | -     |

## OTP\_DATA: KEY3\_VALID Register

**Offset:** 0xf7b

### Description

Valid flag for key 3. Once the valid flag is set, the key can no longer be read or written, and becomes a valid fixed key for protecting OTP pages.

Table 1414.  
KEY3\_VALID Register

| Bits  | Description                                                         | Type | Reset |
|-------|---------------------------------------------------------------------|------|-------|
| 31:17 | Reserved.                                                           | -    | -     |
| 16    | <b>VALID_R2</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 15:9  | Reserved.                                                           | -    | -     |
| 8     | <b>VALID_R1</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 7:1   | Reserved.                                                           | -    | -     |
| 0     | <b>VALID</b>                                                        | RO   | -     |

## OTP\_DATA: KEY4\_VALID Register

**Offset:** 0xf7c

### Description

Valid flag for key 4. Once the valid flag is set, the key can no longer be read or written, and becomes a valid fixed key for protecting OTP pages.

Table 1415.  
KEY4\_VALID Register

| Bits  | Description                                                         | Type | Reset |
|-------|---------------------------------------------------------------------|------|-------|
| 31:17 | Reserved.                                                           | -    | -     |
| 16    | <b>VALID_R2</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 15:9  | Reserved.                                                           | -    | -     |
| 8     | <b>VALID_R1</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 7:1   | Reserved.                                                           | -    | -     |
| 0     | <b>VALID</b>                                                        | RO   | -     |

## OTP\_DATA: KEY5\_VALID Register

**Offset:** 0xf7d

### Description

Valid flag for key 5. Once the valid flag is set, the key can no longer be read or written, and becomes a valid fixed key for protecting OTP pages.

Table 1416.  
KEY5\_VALID Register

| Bits  | Description                                                         | Type | Reset |
|-------|---------------------------------------------------------------------|------|-------|
| 31:17 | Reserved.                                                           | -    | -     |
| 16    | <b>VALID_R2</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 15:9  | Reserved.                                                           | -    | -     |
| 8     | <b>VALID_R1</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 7:1   | Reserved.                                                           | -    | -     |
| 0     | <b>VALID</b>                                                        | RO   | -     |

## OTP\_DATA: KEY6\_VALID Register

**Offset:** 0xf7e

### Description

Valid flag for key 6. Once the valid flag is set, the key can no longer be read or written, and becomes a valid fixed key for protecting OTP pages.

Table 1417.  
KEY6\_VALID Register

| Bits  | Description                                                         | Type | Reset |
|-------|---------------------------------------------------------------------|------|-------|
| 31:17 | Reserved.                                                           | -    | -     |
| 16    | <b>VALID_R2</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 15:9  | Reserved.                                                           | -    | -     |
| 8     | <b>VALID_R1</b> : Redundant copy of VALID, with 3-way majority vote | RO   | -     |
| 7:1   | Reserved.                                                           | -    | -     |
| 0     | <b>VALID</b>                                                        | RO   | -     |

## OTP\_DATA: PAGE0\_LOCK0, PAGE1\_LOCK0, ..., PAGE61\_LOCK0, PAGE62\_LOCK0 Registers

**Offsets:** 0xf80, 0xf82, ..., 0xffa, 0xffc

### Description

Lock configuration LSBs for page  $N$  (rows  $0x40 * N$  through  $0x40 * N + 0x3f$ ). Locks are stored with 3-way majority vote encoding, so that bits can be set independently.

This OTP location is always readable, and is write-protected by its own permissions.

Table 1418.  
PAGE0\_LOCK0,  
PAGE1\_LOCK0, ...,  
PAGE61\_LOCK0,  
PAGE62\_LOCK0  
Registers

| Bits  | Description                                                                                                              | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                                                                                | -    | -     |
| 23:16 | <b>R2</b> : Redundant copy of bits 7:0                                                                                   | RO   | -     |
| 15:8  | <b>R1</b> : Redundant copy of bits 7:0                                                                                   | RO   | -     |
| 7     | Reserved.                                                                                                                | -    | -     |
| 6     | <b>NO_KEY_STATE</b> : State when at least one key is registered for this page and no matching key has been entered.      | RO   | -     |
|       | Enumerated values:                                                                                                       |      |       |
|       | 0x0 → READ_ONLY                                                                                                          |      |       |
|       | 0x1 → INACCESSIBLE                                                                                                       |      |       |
| 5:3   | <b>KEY_R</b> : Index 1-6 of a hardware key which must be entered to grant read access, or 0 if no such key is required.  | RO   | -     |
| 2:0   | <b>KEY_W</b> : Index 1-6 of a hardware key which must be entered to grant write access, or 0 if no such key is required. | RO   | -     |

## OTP\_DATA: PAGE0\_LOCK1, PAGE1\_LOCK1, ..., PAGE61\_LOCK1, PAGE62\_LOCK1 Registers

**Offsets:** 0xf81, 0xf83, ..., 0xffb, 0xffd

### Description

Lock configuration MSBs for page *N* (rows 0x40 \* *N* through 0x40 \* *N* + 0x3f). Locks are stored with 3-way majority vote encoding, so that bits can be set independently.

This OTP location is always readable, and is write-protected by its own permissions.

Table 1419.  
PAGE0\_LOCK1,  
PAGE1\_LOCK1, ...,  
PAGE61\_LOCK1,  
PAGE62\_LOCK1  
Registers

| Bits  | Description                                                                                                                                                                                             | Type | Reset |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                                                                                                                                                               | -    | -     |
| 23:16 | <b>R2</b> : Redundant copy of bits 7:0                                                                                                                                                                  | RO   | -     |
| 15:8  | <b>R1</b> : Redundant copy of bits 7:0                                                                                                                                                                  | RO   | -     |
| 7:6   | Reserved.                                                                                                                                                                                               | -    | -     |
| 5:4   | <b>LOCK_BL</b> : Dummy lock bits reserved for bootloaders (including the RP2350 USB bootloader) to store their own OTP access permissions. No hardware effect, and no corresponding SW_LOCKx registers. | RO   | -     |
|       | Enumerated values:                                                                                                                                                                                      |      |       |
|       | 0x0 → READ_WRITE: Bootloader permits user reads and writes to this page                                                                                                                                 |      |       |
|       | 0x1 → READ_ONLY: Bootloader permits user reads of this page                                                                                                                                             |      |       |
|       | 0x2 → RESERVED: Do not use. Behaves the same as INACCESSIBLE                                                                                                                                            |      |       |
|       | 0x3 → INACCESSIBLE: Bootloader does not permit user access to this page                                                                                                                                 |      |       |

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Type | Reset |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 3:2  | <b>LOCK_NS:</b> Lock state for Non-secure accesses to this page. Thermometer-coded, so lock state can be advanced permanently from any state to any less-permissive state by programming OTP. Software can also advance the lock state temporarily (until next OTP reset) using the SW_LOCKx registers.<br><br>Note that READ_WRITE and READ_ONLY are equivalent in hardware, as the SBPI programming interface is not accessible to Non-secure software. However, Secure software may check these bits to apply write permissions to a Non-secure OTP programming API. | RO   | -     |
|      | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |
|      | 0x0 → READ_WRITE: Page can be read by Non-secure software, and Secure software may permit Non-secure writes.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |       |
|      | 0x1 → READ_ONLY: Page can be read by Non-secure software                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |       |
|      | 0x2 → RESERVED: Do not use. Behaves the same as INACCESSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0x3 → INACCESSIBLE: Page can not be accessed by Non-secure software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |       |
| 1:0  | <b>LOCK_S:</b> Lock state for Secure accesses to this page. Thermometer-coded, so lock state can be advanced permanently from any state to any less-permissive state by programming OTP. Software can also advance the lock state temporarily (until next OTP reset) using the SW_LOCKx registers.                                                                                                                                                                                                                                                                      | RO   | -     |
|      | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |       |
|      | 0x0 → READ_WRITE: Page is fully accessible by Secure software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |       |
|      | 0x1 → READ_ONLY: Page can be read by Secure software, but can not be written.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0x2 → RESERVED: Do not use. Behaves the same as INACCESSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |       |
|      | 0x3 → INACCESSIBLE: Page can not be accessed by Secure software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |       |

## OTP\_DATA: PAGE63\_LOCK0 Register

Offset: 0xffe

### Description

Lock configuration LSBs for page 63 (rows 0xfc0 through 0xfff). Locks are stored with 3-way majority vote encoding, so that bits can be set independently.

This OTP location is always readable, and is write-protected by its own permissions.

Table 1420.  
PAGE63\_LOCK0  
Register

| Bits  | Description                                                                                                                                                                       | Type | Reset |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                                                                                                                                         | -    | -     |
| 23:16 | <b>R2:</b> Redundant copy of bits 7:0                                                                                                                                             | RO   | -     |
| 15:8  | <b>R1:</b> Redundant copy of bits 7:0                                                                                                                                             | RO   | -     |
| 7     | <b>RMA:</b> Decommission for RMA of a suspected faulty device. This re-enables the factory test JTAG interface, and makes pages 3 through 61 of the OTP permanently inaccessible. | RO   | -     |
| 6     | <b>NO_KEY_STATE:</b> State when at least one key is registered for this page and no matching key has been entered.                                                                | RO   | -     |
|       | Enumerated values:                                                                                                                                                                |      |       |

| Bits | Description                                                                                                              | Type | Reset |
|------|--------------------------------------------------------------------------------------------------------------------------|------|-------|
|      | 0x0 → READ_ONLY                                                                                                          |      |       |
|      | 0x1 → INACCESSIBLE                                                                                                       |      |       |
| 5:3  | <b>KEY_R</b> : Index 1-6 of a hardware key which must be entered to grant read access, or 0 if no such key is required.  | RO   | -     |
| 2:0  | <b>KEY_W</b> : Index 1-6 of a hardware key which must be entered to grant write access, or 0 if no such key is required. | RO   | -     |

## OTP\_DATA: PAGE63\_LOCK1 Register

Offset: 0xfff

### Description

Lock configuration MSBs for page 63 (rows 0xfc0 through 0xfff). Locks are stored with 3-way majority vote encoding, so that bits can be set independently.

This OTP location is always readable, and is write-protected by its own permissions.

Table 1421.  
PAGE63\_LOCK1  
Register

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Type | Reset |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 31:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -    | -     |
| 23:16 | <b>R2</b> : Redundant copy of bits 7:0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | RO   | -     |
| 15:8  | <b>R1</b> : Redundant copy of bits 7:0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | RO   | -     |
| 7:6   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -    | -     |
| 5:4   | <b>LOCK_BL</b> : Dummy lock bits reserved for bootloaders (including the RP2350 USB bootloader) to store their own OTP access permissions. No hardware effect, and no corresponding SW_LOCKx registers.                                                                                                                                                                                                                                                                                                                                                                  | RO   | -     |
|       | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |       |
|       | 0x0 → READ_WRITE: Bootloader permits user reads and writes to this page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |       |
|       | 0x1 → READ_ONLY: Bootloader permits user reads of this page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |       |
|       | 0x2 → RESERVED: Do not use. Behaves the same as INACCESSIBLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |       |
|       | 0x3 → INACCESSIBLE: Bootloader does not permit user access to this page                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |       |
| 3:2   | <b>LOCK_NS</b> : Lock state for Non-secure accesses to this page. Thermometer-coded, so lock state can be advanced permanently from any state to any less-permissive state by programming OTP. Software can also advance the lock state temporarily (until next OTP reset) using the SW_LOCKx registers.<br><br>Note that READ_WRITE and READ_ONLY are equivalent in hardware, as the SBPI programming interface is not accessible to Non-secure software. However, Secure software may check these bits to apply write permissions to a Non-secure OTP programming API. | RO   | -     |
|       | Enumerated values:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |       |
|       | 0x0 → READ_WRITE: Page can be read by Non-secure software, and Secure software may permit Non-secure writes.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |      |       |
|       | 0x1 → READ_ONLY: Page can be read by Non-secure software                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |      |       |
|       | 0x2 → RESERVED: Do not use. Behaves the same as INACCESSIBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |      |       |
|       | 0x3 → INACCESSIBLE: Page can not be accessed by Non-secure software.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |       |

| Bits | Description                                                                                                                                                                                                                                                                                         | Type | Reset |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| 1:0  | <b>LOCK_S</b> : Lock state for Secure accesses to this page. Thermometer-coded, so lock state can be advanced permanently from any state to any less-permissive state by programming OTP. Software can also advance the lock state temporarily (until next OTP reset) using the SW_LOCKx registers. | RO   | -     |
|      | Enumerated values:                                                                                                                                                                                                                                                                                  |      |       |
|      | 0x0 → READ_WRITE: Page is fully accessible by Secure software.                                                                                                                                                                                                                                      |      |       |
|      | 0x1 → READ_ONLY: Page can be read by Secure software, but can not be written.                                                                                                                                                                                                                       |      |       |
|      | 0x2 → RESERVED: Do not use. Behaves the same as INACCESSIBLE.                                                                                                                                                                                                                                       |      |       |
|      | 0x3 → INACCESSIBLE: Page can not be accessed by Secure software.                                                                                                                                                                                                                                    |      |       |